

The Pension and the Drawdown Rule

Why a Means Test Makes the Portfolio and the Withdrawal Rule One Decision

This version: 13 September 2026

Abstract

An all-equity lifecycle portfolio is said to dominate the age-declining glide path of a target-date fund. We show that under a means-tested public pension this is not a property of the pension alone. The test assesses what the portfolio still holds, so whether a good return reaches the retiree depends on whether their withdrawal rule spends it or leaves it to be assessed — and the ordering that follows is a joint property of the pension, the withdrawal rule and the household's risk aversion, none of which is sufficient alone. On a 16-country panel of real returns spanning 1890–2020, crossing five public pension regimes with eight withdrawal rules on common simulated lifetimes, replacing an earnings-related pension with a means-tested one paid to the same household on the same contributions reverses the portfolio ordering under one of the eight rules and under no other: the lead runs -13.06% under the fixed real withdrawal the literature assumes, which never spends what the portfolio earns, and +11.04% to +21.09% under the seven rules that do. Against an earnings-related pension the same household leads by +11.09% whatever rule it spends by. The reversal's sign holds in all sixteen delete-one-country sub-panels, and it is absent at the lowest of the three risk aversions we score: re-read at $\gamma = 2$ the same cell is +10.71%, and the all-equity portfolio leads again, so the third condition binds as hard as the other two. The mechanism is the loss of an unconditional floor rather than the means test's taper, and either of the two institutions a country can change — a compulsory contribution, worth +11.0 points here, or a withdrawal rule that cannot deplete, worth up to 34 points — offsets most of it. Near the test, the drawdown default and the portfolio default are one decision.

What the paper shows. (i) Under a means-tested pension the portfolio ordering is a property of the withdrawal rule: replacing an earnings-related pension with a means-tested one, holding contributions fixed, reverses it under the rule the literature assumes and under no other rule in an eight-rule menu, and the rule that reverses it is the only rule in that menu whose payment never reads the balance. The reversal needs a household risk-averse enough to pay for a floor as well, so the interaction is three-way and not two. (ii) The mechanism is the floor, not the taper: paying the *same* flat pension to the same household without testing it leaves the all-equity portfolio ahead by +12.78% against -13.06% with the test applied — the rate, the contributions and the panel all held — and 83% of those households are past the cut-off before the test applies, so the taper reaches them only in the left tail. (iii) A withdrawal rule that cannot deplete restores the ordering: on the same household, under the rule Section 8 selects, the lead moves from -13.06% to +19.69%, and every rate in the family restores it. (iv) Sixteen delete-one-country sub-panels sign the reversal at matched contributions and sign the rule effect under both pensions; what they cannot sign is the net of the two institutions when a country moves both at once. Resampling the panel with replacement rather than deleting from it — which drops the independence a jackknife assumes — gives intervals within a fifth of the delete-one ones and changes no sign. (v) Who pays for compulsory saving changes lifetime consumption by 13.4% and the retiree's problem not at all. (vi) On a household the test does bind, the wanted equity share is a property of the withdrawal rule rather than of the pension — 0% under a fixed real withdrawal and 100% under a percentage of balance, in every one of twelve cells crossing homeowners with renters, singles with couples, and all three positions against each household's own thresholds.

Keywords: lifecycle asset allocation; public pension design; means testing; target-date funds; certainty-equivalent consumption; decumulation. **JEL:** G11, G51, H55, D14.

1. Introduction

The default investment vehicle of the modern retirement system is the target-date fund, and it embodies one proposition: that an investor should hold less equity as they age. Anarkulova, Cederburg and O'Doherty (2023) challenge it directly, showing that when returns are drawn in blocks from the international historical record rather than i.i.d. from post-war US data, a fixed all-equity portfolio beats the glide path on almost every metric an investor would care about. The finding has been influential precisely because it is robust to the things a reader first reaches for: it survives higher risk aversion, a longer horizon, and a return panel that includes every disaster the twentieth century supplied.

It is not robust to the thing nobody reaches for, which is the pension. That result is derived for an investor who receives the United States' Social Security: an earnings-related benefit, paid whatever else the retiree owns. Roughly a third of OECD members instead pay a residence-based benefit that is withdrawn against the retiree's own assets or income, Australia's Age Pension among them. This paper asks what the all-equity prescription becomes for an investor under that second kind of scheme, and finds that it reverses.

The exercise is deliberately narrow. Nothing about the return process changes between our two arms: the same panel, the same bootstrap, the same simulated lifetimes, the same household, the same fund menu. Only the benefit formula moves. Whatever difference appears is therefore attributable to the pension and to nothing else, which is what lets the paper say *why* the prescription reverses rather than only *that* it does.

Two households appear below and it is worth separating them here rather than leaving a reader to notice.

The first saves this paper's own 10% of income and is the household every comparative statement is made on, because holding the contribution rate still is what lets a difference be attributed to the pension. The second adds the compulsory Superannuation Guarantee the calibration is drawn from, for a total of 20.2%, and is the household the legislated combination produces. They are the same worker with two savings rates, and they arrive at retirement in different places: 13.8 times average earnings against 32.1, with 83% and 98% of them respectively past the point where the assets test stops paying.

That distance matters for the mechanism, and it is the reason to lead with the first. Even the 10% saver is mostly past the cut-off before the test is applied — the pension replaces 3.9% of their career income against the earnings-related schedule's 43%, and 1.9% of the mandated saver's — so neither headline is mainly a comparison of two means-test designs. Both are comparisons between a household that receives an unconditional pension and the same household receiving almost none, which is exactly the intervention the mechanism section says should matter and the one the taper reading says should not. The households a means test actually binds are a third population, and Section 7 reaches them by scaling the balance rather than by pretending either of these is among them.

1.1 What we find

Under a means-tested pension the portfolio ordering is a property of the withdrawal rule. A means test assesses what the portfolio still holds, so whether a good return helps the retiree depends on whether their rule spends it or leaves it to be assessed. Run the same means-tested pension against the whole menu and the all-equity portfolio leads the target-date fund under seven of the eight withdrawal rules, by +11.04% to +21.09%. It loses under one, and that one is the fixed real withdrawal the literature assumes and this paper's earlier sections spend by — the only rule *in this menu* whose payment never reads the balance. The menu is eight rules and not a taxonomy: what the grid establishes is a division, not that no other rule could fall on the losing side of it.

Table 1. The all-equity portfolio's lead over the target-date fund under a means-tested pension, rule by rule, at a matched contribution rate.

Withdrawal rule	All-equity lead	Reads the balance?
fixed real	-13.06%	no
amortisation at 0%	+11.04%	yes
percentage of balance at 4%	+13.26%	yes
amortisation at 2%	+13.68%	yes
amortisation at 4%	+16.90%	yes
amortisation at 6%	+19.69%	yes
amortisation at 10%	+20.84%	yes
amortisation at 8%	+21.09%	yes

Notes. The cell the panel resolves: the same household, the same contributions and the same simulated lifetimes in every row, with only the withdrawal rule changing. “Reads the balance” marks a rule whose payment in a given year is a function of the portfolio that year. Section 9 gives the full grid, the delete-one-country intervals and the other four pension regimes.

The control is what makes that an interaction rather than a ranking of rules. Run the same eight rules against a pension that is not tested and none of them reverses anything: the eight run +12.78% to +23.69% under the same flat pension without the test and +11.09% to +20.24% under an earnings-related pension, every cell positive. The withdrawal rule is worth five to nine points of certainty-equivalent consumption whatever pension the household retires onto — that much is just a better rule — and it changes the *sign* of the portfolio comparison in one place only, which is where the test operates.

That is the paper's result, and the rest of it is an account of why. The reversal below is the same fact stated on the one rule it happens under; we lead with it because it is what the literature assumes, not because it is what a retiree should do. Section 8 solves for the rule a retiree should actually use, and it is not that one.

On that rule, the reversal is as clean as a pair of runs differing in nothing else. Pay this household the calibrated flat rate with no test attached and the all-equity portfolio leads the target-date fund by +12.78%; apply the test to that identical benefit and the lead becomes -13.06%, and the de-risking glide path takes first place. The rate, the contribution rate, the panel and the simulated lifetimes are the same in both; only the test is switched on. Against the earnings-related schedule instead, the same household leads by +11.09%, which is the anchor to the literature and bundles the level change with the test — worth +1.69 points, so nothing in what follows turns on which of the two references is used.

That sign is one the cross-section resolves. All sixteen of the fifteen-country sub-panels put the target-date fund ahead, and the least negative of them is -8.42% — clear of zero rather than close to it. The delete-one interval, [-22.02, -4.11], agrees, and we lead with the count rather than the interval for a reason Section 9.3 gives: this cell's deletions are skewed enough that the standard error built on them is the weaker of the two statistics.

The combination one country legislates is a different and smaller number, because a second institution pushes back. The Superannuation Guarantee doubles the contribution rate, and under a means-tested pension that is worth +11.0 points — a bigger balance is a partial substitute for the floor the test removed. Run together the two land at -2.08%, and that residual is the one cell of the table this cross-section cannot sign. We report both, in that order: the first is a statement about pension design and is what the mechanism sections are about; the second is a statement about one country's particular combination of two, and is the more fragile of the two.

One further qualification belongs with those sentences rather than after them. The legislated cell is the one this cross-section cannot resolve: a delete-one-country jackknife over the sixteen markets puts a standard error of 10.3 points on a point estimate of -2.08%, and the sign changes when eight of the sixteen are removed one at a time. Every other cell in that table has a standard error between 2 and 5 points and an interval excluding zero.

A third condition is the risk aversion. Re-scoring the same simulated lifetimes at risk aversions of 2, 5 and 10, the matched means-tested cell runs +10.71% at $\gamma = 2$, -13.06% at $\gamma = 5$ and -33.66% at $\gamma = 10$, and the legislated one runs +21.91% at $\gamma = 2$, -2.08% at $\gamma = 5$ and -36.40% at $\gamma = 10$: in both the reversal is absent at the lowest of the three and deepens through the other two, while every other withdrawal rule stays positive at all three. So the failure of the all-equity prescription is located precisely — an asset-tested pension, a fixed real withdrawal, and a household risk-averse enough to pay for the floor — and two of those three are chosen by whoever sets the defaults.

The mechanism is the floor, not the taper. A means test is usually discussed as an implicit tax on wealth, and Australia's is steep enough to qualify — a dollar of assessable assets costs 7.8% of pension a year, more than domestic equity earns on average in this panel. But that reading does not survive checking where the household stands. 83% of the matched households and 98% of the mandated ones are past the cut-off before the test is applied, so the taper reaches either only in the left tail, after a portfolio has already fallen. What the means test removes is the unconditional annuity itself. A retiree standing on a guaranteed floor can carry the equity tail; one standing on their portfolio alone cannot, and a risk-averse objective prices that difference heavily.

A floor need not come from a pension. The reversal is conditional on a withdrawal rule that can run out. Under an amortisation rule the balance is divided by the years remaining, ruin falls to zero, and the household manufactures its own floor out of the portfolio it already holds. Holding the pension, the contributions and the returns fixed and changing only the rule, the all-equity lead on the household the reversal is reported on moves from -13.06% to +19.69% under the amortisation at 6% rule Section 8 selects — and to +21.09% at the most generous rate in the family — so the portfolio ordering reverses a second time. The all-equity prescription is therefore conditional on two institutions rather than one — and, as the next paragraph but one reports, on a preference as well.

It is worth being exact about what that sentence compares, because two orderings are easy to run together. One is between *portfolios*: which of two funds a given retiree should hold. The other is between *countries*: which pension system delivers more retirement consumption. They can move in opposite directions, and the claim above is about the first. Every percentage in this paper is a portfolio comparison unless it is labelled otherwise.

Who pays for compulsory saving changes the level and not the portfolio. The Superannuation Guarantee the mandated arms are calibrated to is levied on the employer, and modelling it that way hands those arms income the voluntary ones never receive. Charging it to wages instead costs 13.4% of lifetime certainty-equivalent consumption — so every comparison here is bracketed by that figure — but it leaves the balance at the pension age, and therefore the entire retirement problem, exactly unchanged. The contribution is made either way. That is an identity rather than an estimate, and it means the incidence question and the portfolio question are separate questions.

On a household the test actually binds, the portfolio is a property of the withdrawal rule, not of the pension. The two households above retire on 13.8 and 32.1 times average earnings, both far outside the taper band, so we scale the arriving balance until it sits inside and ask again. Spending a fixed real amount, the household wants no equity at any balance the test can reach; spending a share of the current balance, it wants at least 100% equity at every one of them, and more than that wherever it can borrow. The reason is mechanical and, once seen, obvious: under a fixed real rule a good return is never spent, so it accumulates into assessable assets, the pension is withdrawn against them at 7.8% a year, and consumption does not rise at all. The upside is confiscated and the downside is not.

1.2 Contribution

Three things here are new, and it is worth separating them from what is replication. That an all-equity portfolio beats a glide path on an international block bootstrap is Anarkulova, Cederburg and O'Doherty's result, and Section 5 reproduces it rather than claiming it.

First, the *conditionality*: the prescription is a property of the pension the investor retires onto, and the design says which feature of the pension it is. Paying the same flat rate to the same household on the same contributions, but without testing it, leaves the all-equity portfolio ahead by +12.78%; applying the test to that identical benefit takes it to -13.06%. Withdrawing a pension at the margin and paying a smaller one are different interventions, and only the first reorders portfolios. Second, the *mechanism*: Section 2 writes the retiree's budget line as three regimes and shows that inside the taper band a means test is insurance rather than a wealth tax. That prediction is sharp enough to fail, and it does: under a fixed real withdrawal the retiree wants no equity anywhere the test reaches, and under a percentage-of-balance rule — once the grid is widened far enough to have an opinion — the maximum sits below the free area rather than inside the band. The taper does insure, and it insures less than the guarantee it withdraws. We report the refutation because a mechanism that cannot be wrong is not a mechanism. Third, the *interaction*: the portfolio and the drawdown rule are one decision when the pension is asset-tested, and we measure how much of one.

1.3 Relation to the literature

The lifecycle portfolio-choice literature since Cocco, Gomes and Maenhout (2005) and Gomes and Michaelides (2005) treats public pension income as a bond-like endowment that crowds fixed income out of the financial portfolio. That is right for an earnings-related schedule. It is the wrong intuition for an asset-tested one, where the endowment is withdrawn precisely as the portfolio grows, and where — as Hubbard, Skinner and Zeldes (1995) show for saving — the binding feature is the floor the transfer provides rather than its expected level. Dahlquist, Setty and Vestman (2018) solve for a default fund's allocation given a pension system; we vary the system and hold the fund menu fixed, which isolates the same interaction from the other side.

The Australian assets test has its own literature, and it reaches the pension-as-endowment question from the institutional side. Iskhakov, Thorp and Bateman (2015) solve an Australian retiree's annuity purchase with the means test in the constraint set, and find the test reshaping a decision that is about buying a floor — which is the same margin this paper reaches through the withdrawal rule, with the annuity itself left out of the choice set. Bateman et al. (2018) document what Australian members actually hold. On the theory side, Sefton, van de Ven and Weale (2008) and Braun, Kopecky and Koreshkova (2017) analyse means-tested transfers as insurance rather than as taxes, and Bütler, Peijnenburg and Staubli (2017) show a means-tested benefit crowding out voluntary annuitisation for exactly the reason the floor matters here. What we add to that work is the comparative statement — the same household, the same returns, two pension systems — and the interaction with the drawdown rule, which we have not found measured.

On the decumulation side the rule that supplies the floor is the actuarial or amortisation rule of Milevsky and Huang (2011). The withdrawal-rule literature typically conditions on a fixed income floor and the pension literature on a fixed withdrawal rule, which is why the interaction between them has room to be new.

The means-testing literature itself is largely about saving and labour supply rather than portfolio choice, and it has long understood the taper as an implicit tax. Hubbard, Skinner and Zeldes (1995) is the closest antecedent to what we find: their asset-tested transfer discourages saving not because of its expected level but because of the floor it puts under consumption. We measure the same feature acting on the composition of the portfolio rather than its size, and find that it acts in the opposite direction to the tax intuition — the taper, inside the band, makes equity cheaper rather than dearer.

Our panel is the Jordà–Schularick–Taylor macrohistory database, the same source the replicated study uses, and the same 16 developed markets Dimson, Marsh and Staunton (2002) cover for a comparable period. Every number below is regenerated from that panel by the pipeline described in Section 4; the fuller robustness apparatus, and the extensions not needed for this argument, are in the companion study.

1.4 Roadmap

Section 2 sets out the retiree's budget line under an asset test and derives the prediction the rest of the paper tests. Sections 3 and 4 describe the panel and the simulation. Section 5 reproduces the all-equity result under the earnings-related schedule, and Section 6 sets out the pension regimes it is scored under. Section 7 charges the guarantee and moves the household onto the test; Section 8 finds the withdrawal rule a retiree should actually use, and Section 9 asks what that rule does to the portfolio ordering and how precisely sixteen countries can resolve each answer. Section 10 says what would change these conclusions. The two systems also differ in their tax treatment, and Section 34 of the companion study rules that out as an explanation; we do not repeat it here.

2. What an Asset Test Does to a Retiree's Budget Line

Write W for the assessable balance a household holds at the start of a retirement year, R for the gross real return it earns over that year, x for what its withdrawal rule pays out, A for the assets-test free area, b for the full pension and τ for the rate at which the pension is withdrawn against assets above A . The test is assessed on the assets that remain once the year's withdrawal has been taken, which is the convention the Australian scheme uses and the one the simulation implements. Consumption in the year is the withdrawal plus whatever pension the test leaves:

$$c = x + \min\{b, \max[0, b - \tau(WR - x - A)]\} \quad (1)$$

The three regimes are the three branches of that expression, and writing them out is what makes the mechanism visible. Let $S = WR - x$ be the assets the test sees. Below the free area ($S \leq A$) the pension is paid in full; inside the band ($A < S < A + b/\tau$) it is withdrawn at the taper; above the cut-off it is gone:

$$c = x + b \quad (\text{below the free area}) \quad (2)$$

$$c = x(1 + \tau) - \tau WR + (b + \tau A) \quad (\text{inside the taper band}) \quad (3)$$

$$c = x \quad (\text{above the cut-off}) \quad (4)$$

2.1 Why the withdrawal rule decides the sign

A means test is usually described as an implicit tax on wealth, and Australia's is steep enough to qualify: at 7.8% a year it exceeds what domestic equity earns on average in this panel. The middle line says that description is incomplete, and what it puts in its place depends entirely on how x responds to R . Differentiate it with respect to the return:

$$\partial c / \partial R = (1 + \tau) \partial x / \partial R - \tau W \quad (5)$$

Take the rule that spends a fixed real amount first, because it is the one the literature assumes and the one this paper spends by. Here x is a constant, so $\partial x / \partial R = 0$ and the derivative is $-\tau W$, negative at every withdrawal rate. A good return raises assessable assets and leaves the withdrawal exactly where it was — this year and every year after it, because the rule does not read the balance at all. The gain is taxed at 7.8% a year for as long as it is held and reaches consumption in no year at all; the only place it can surface is the estate. The upside is confiscated and the downside is not.

Now the rule that spends a fixed share k of the current balance. Here $x = kWR$, so $\partial x/\partial R = kW$ and the derivative is $W[k(1 + \tau) - \tau]$. This is positive only above a threshold withdrawal rate, $k > \tau/(1 + \tau)$, which on the Australian taper is 7.24%. It is worth being blunt that the proportional rule this paper actually simulates draws 4% and therefore sits *below* that threshold: in the year a good return arrives, this household too hands the test more in withdrawn pension than it takes in extra spending.

The two rules nonetheless differ in the way that matters, and the threshold is what makes the difference visible rather than what creates it. Under the proportional rule the gain is deferred: the larger balance is still there, and every later withdrawal is a share of it, so the return reaches consumption eventually whatever k is. Under the fixed real rule the deferral never ends, because no future withdrawal reads the balance either. A proportional rule spends the gain late; a fixed real rule never spends it. That is the distinction which survives from one period to a retirement, and the threshold only says how late.

Proposition 1 (return exposure under a means test is governed by the withdrawal rule). Let the household hold assessable balance W , earn gross real return R , and withdraw x . Inside the taper band, where the pension is being withdrawn at τ per unit of assessable assets:

- **(i)** Consumption falls in the return — $\partial c/\partial R < 0$ — if and only if $\partial x/\partial R < \tau W/(1 + \tau)$. The threshold is a property of the withdrawal rule; the taper only sets where it sits.
- **(ii)** A rule whose payment does not read the balance has $\partial x/\partial R = 0$ and therefore satisfies (i) at every balance and every withdrawal rate. The return it forgoes is never recovered, because no later payment reads the balance either: the gain is assessed for as long as it is held and reaches consumption in no year.
- **(iii)** A rule paying a share $k > 0$ of the current balance has $\partial x/\partial R = kW$ and violates (i) exactly when $k > \tau/(1 + \tau)$. Below that threshold it satisfies (i) in the year the return arrives, but the larger balance is carried forward and every later payment is a share of it, so the return reaches consumption in some later year for any $k > 0$.

The content of the proposition is (ii) against (iii). The taper sets where the threshold sits — it is $\tau W/(1 + \tau)$, and without a test there is no threshold at all — but which side of it a household falls on is decided by $\partial x/\partial R$, and the two rule families sit on opposite sides regardless of the rate either is run at. What the proposition does *not* deliver is the optimal risky share. One period of accounting signs a derivative; the share depends on the return distribution, the horizon and the curvature of the objective, and none of those appear above. Section 7 measures the share directly and Section 9 measures what it does to the portfolio ordering; the proposition says which way to expect them to come out, and Section 2.2 sets out where it can be wrong.

Corollary 1. Take a rule with $\partial x/\partial R = 0$, and a retirement year in which the portfolio funds its withdrawal. Consumption is then $x + b$ below the free area and x above the cut-off, neither of which moves with the return, and $x(1 + \tau) - \tau WR + (b + \tau A)$ inside the band, which falls in it. So at no position does a better return raise that year's consumption, and inside the band it lowers it — and, since the balance carried forward is larger, it lowers the following year's too for as long as the household remains in the band. The risky asset contributes nothing to consumption utility and subtracts from it wherever the test operates. It enters the objective positively only through the estate.

That is a statement about the budget line and not about the investor, and it is the reason to expect a corner rather than a small allocation: no risk aversion, no return distribution and no horizon appears anywhere in it, so nothing in the calibration is available to overturn it. Section 7.4 is the test — the corner holds at half the baseline risk aversion, at twice it, and under two consumption floors — and the corollary is why it should. One channel is outside the algebra and belongs with it: a good return postpones depletion, and a rule that spends a fixed real amount can deplete. On the paths where it does, the return does reach consumption, by buying years in which there is a withdrawal at all. Section 9.2 reports how many paths those are.

2.1.1 What that implies for the portfolio

The three regimes then order the retiree's problem, and the comparison is two lines. Exposure to the return is $\partial x/\partial R$ below the free area, $(1 + \tau)\partial x/\partial R - \tau W$ inside the band, and $\partial x/\partial R$ again above the cut-off. The part of consumption that does not move with the return goes b , then $b + \tau A$, then nothing. So inside the band the household faces strictly less exposure than below the free area and carries a strictly larger riskless component; above the cut-off it faces the same exposure as below the free area and carries none at all.

Two words are worth avoiding here, and the reason is a real objection rather than fastidiousness. It is tempting to call $b + \tau A$ a *floor*, and it is not one: consumption inside the band is *decreasing* in the return, so that quantity is the intercept of a downward-sloping line and is attained at the top of the band rather than guaranteed across it. What the algebra delivers is the riskless component of consumption, which is the quantity the endowment result is about and which does rise. Elsewhere in this paper “floor” means the unconditional pension — a payment that arrives whatever the portfolio does — and that is a floor in the ordinary sense.

For an investor with constant relative risk aversion, a larger riskless component of consumption raises the optimal share of the risky asset — the standard result for a bond-like endowment, and the one the lifecycle literature invokes to explain why a pension crowds fixed income out of the financial portfolio. Applying it to the ordering above gives the prediction directly: **optimal equity should be highest inside the taper band, lower below the free area, and lowest above the cut-off** — non-monotone in wealth, with its minimum where the pension has been withdrawn entirely rather than where it is being withdrawn.

That is the prediction under a rule that spends the balance. Under a fixed real rule the insurance term never arrives, because the household never converts the return into anything it eats, and the prediction is a corner rather than a shape: no equity anywhere the taper operates, and position against the test should barely matter, because what does the damage is the rule and not the band.

On this calibration the free area is 3.01 times average earnings, the full rate 29.3% of them, and the cut-off 6.77. Section 7 runs both rules across all three regimes and reports what happens to each prediction. Neither survives intact, and saying so is the point of writing one down.

2.2 What the model does not settle

This is one period of accounting, not a solved lifecycle problem. It says which way the forces point; it does not say how large the non-monotonicity is, or whether it is there at all, because that depends on the return distribution, the horizon and the risk aversion. Section 7.5 finds it is not there: under either withdrawal rule wanted equity turns out to be monotone in wealth, and the hump this section predicts does not appear once the grid is wide enough to show one. Nor does one period describe the region a real means test spends most of its time in, where a household inside the band this year is over the cut-off next year and the test is re-assessed annually against a drawn-down balance. Both are simulation questions, and the simulation is the rest of the paper.

The dependence on risk aversion turns out to be the sharper of those two omissions, and worth flagging here rather than letting it arrive as a surprise. One period of accounting says the curvature of the objective scales the effect; Section 9.5 finds that over the range of risk aversions commonly used it changes the effect's *sign*. Nothing in the algebra above rules that out — the floor and the exposure move in opposite directions across the band, and which dominates is exactly the question a felicity function answers — but the algebra does not predict it either, and we did not expect it.

One caution belongs with the second prediction rather than with the results. It is a corner at zero, and a constant-relative-risk-aversion objective with a consumption floor near zero is unbounded below — a handful of near-starvation years can move a certainty equivalent further than a decade of ordinary ones. A prediction of “no

equity” is exactly the kind one should distrust, so Section 7 re-scores it at several risk aversions and several consumption floors before reporting it.

3. Data

Everything below rests on one panel, and the argument of this paper is a comparison of two objective functions over it rather than a claim about the returns themselves. So this section is kept to what a reader needs to judge that comparison: what the panel is, which markets are in it, and what is known to be wrong with it. The construction detail, the source audit and the coverage arithmetic are in the companion study.

3.1 What the panel is

The panel is an annual, country-by-year matrix of five real series — domestic equity, international equity, long-term government bonds, short-term bills and consumer price inflation — for 16 developed markets over 1890–2020. It contains 2,010 usable country-years of domestic equity returns, with a median country history of 125 years and a minimum of 115 years.

All series are *real*: nominal total returns are deflated by each country's own consumer price index in the same year, so that a hyperinflation shows up as a catastrophic real return rather than as a spectacular nominal one. This is not a cosmetic choice. Several of the worst episodes in the sample — Germany in the early 1920s, central Europe in the late 1940s — are invisible in nominal space and dominate the left tail in real space.

The equity series across the panel average 6.8% in arithmetic real terms and 4.3% geometrically, with a mean cross-country standard deviation of 22.2%. The gap between the arithmetic and geometric means — over three percentage points — is itself the story: at this level of volatility, the compounded experience of a lifetime investor is far below the average annual return, and any simulation that draws i.i.d. from the arithmetic mean will overstate what a real investor would have received.

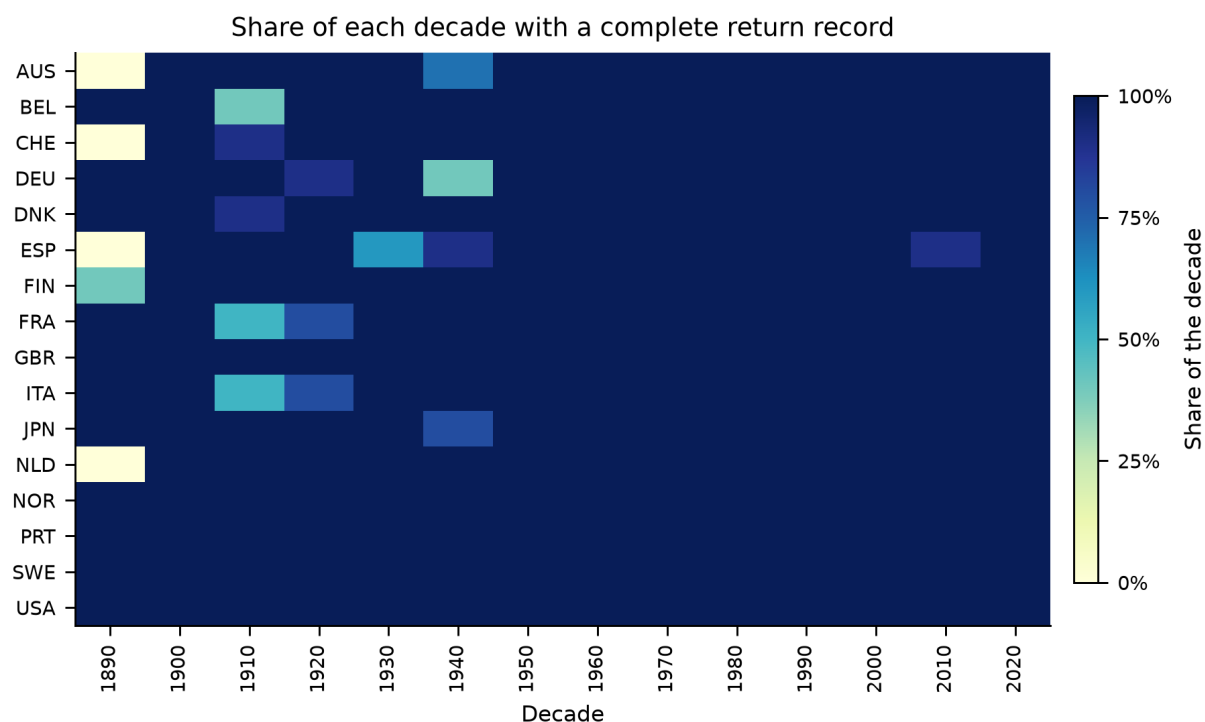


Figure 1.

Share of each decade for which a country has a complete return record. Darker is more complete; the scale is a share so that the final, partial decade compares with the rest. The gaps are not random: they cluster around the two world wars and around the market closures that follow them, which is precisely the period a survivorship-prone sample would drop.

3.2 The country set

The Jordà–Schularick–Taylor database carries complete, independently sourced equity, bond and bill total returns for 16 countries: Australia, Belgium, Denmark, Finland, France, Germany, Italy, Japan, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, United States. That set is the panel, and every country-year of every return series in it is a recorded observation.

The wider developed-market universe — IMF advanced economies with an investable domestic equity market, plus Poland — numbers 38. The histories of the other 22 exist in commercial databases such as Global Financial Data and Dimson–Marsh–Staunton, which are proprietary and not redistributable. Working from openly licensed sources, 16 is the extent of the recorded evidence, and Section 3.6.1 of the companion study reports what the excluded countries do carry.

That breadth is the paper's principal limitation, and Section 10.1 develops it. Its most direct consequence is on the international leg, which is a leave-one-out average and therefore spans 15 foreign markets, all advanced economies with long histories. Every statement here about international diversification is a statement about that set.

3.3 Constructing the international leg

An investor in country i holding "international equity" does not hold a global index that includes their own market. For each country-year we therefore build the international return as a leave-one-out average across all other countries with data that year, computed in gross terms and in a common currency before being converted back to the home investor's real terms. Leaving the home market out is what makes the domestic and international legs genuinely distinct assets rather than two overlapping slices of the same one.

Three observations in the raw international series are extreme enough to dominate any lifetime that draws them, all of them artefacts of a market reopening after wartime closure with a collapsed price index. We winsorise the international leg at the 0.1th percentile of its own distribution; the affected observations are listed in full in Table 4 so that the reader can judge the intervention rather than take it on trust.

Table 2. Real domestic equity returns by country

Country	Years	From	To	Mean	Geo. mean	S.d.	Skew	AR(1)
Australia	118	1900	2020	7.8%	6.4%	16.8%	-0.31	-0.07
Belgium	125	1890	2020	5.9%	3.8%	20.7%	-0.00	0.08
Switzerland	120	1900	2020	6.8%	5.1%	18.9%	0.30	0.10
Germany	124	1890	2020	8.7%	4.6%	28.9%	1.08	-0.01
Denmark	130	1890	2020	7.5%	6.1%	17.2%	0.43	0.04
Spain	115	1900	2020	6.2%	4.2%	21.1%	0.77	0.32
Finland	125	1896	2020	8.9%	5.0%	30.0%	1.21	0.18
France	124	1890	2020	3.6%	1.1%	23.3%	0.86	0.15
United Kingdom	131	1890	2020	6.6%	4.9%	18.7%	0.90	-0.05
Italy	124	1890	2020	6.1%	2.6%	26.8%	0.66	-0.04
Japan	129	1890	2020	7.9%	4.2%	25.8%	0.26	0.12
Netherlands	121	1900	2020	7.1%	5.1%	21.2%	0.74	0.03
Norway	131	1890	2020	5.7%	3.7%	20.1%	0.46	0.02
Portugal	131	1890	2020	3.2%	-0.1%	26.0%	0.66	0.25
Sweden	131	1890	2020	8.1%	6.1%	20.3%	0.05	0.13
United States	131	1890	2020	8.4%	6.7%	18.7%	-0.24	0.00

Notes. Arithmetic and geometric means are annual real returns deflated by each country's own consumer price index; AR(1) is the first-order autocorrelation of the annual series. Appendix B of the companion study repeats this with excess kurtosis.

3.4 Cross-asset structure

The correlation structure the bootstrap must preserve is reported below. Domestic and international equity correlate at 0.43, which is high enough that international diversification is not free and low enough that it is worth having. Bonds and bills correlate at 0.72. Inflation correlates negatively with every asset, most strongly with bills (-0.66), which is the mechanism by which cash loses in real terms over long horizons.

Table 3. Pooled cross-asset correlation matrix of annual real returns

	Dom. equity	Intl. equity	Bonds	Bills	Inflation
Domestic equity	1.000	0.429	0.315	0.234	-0.121
International equity	0.429	1.000	0.110	0.117	-0.042
Bonds	0.315	0.110	1.000	0.716	-0.469
Bills	0.234	0.117	0.716	1.000	-0.662
Inflation	-0.121	-0.042	-0.469	-0.662	1.000

Notes. Pooled across all country-years in the panel. These are the moments the block bootstrap is required to reproduce; Table 5 reports how closely it does so.

Table 4. Every winsorised observation in the international leg

Year	Country	Raw return	After winsorisation
1942	Italy	1120%	301%
1945	Japan	-86%	-70%
1946	France	308%	301%

Notes. All three are markets reopening after wartime closure against a collapsed price index. Left unwinsorised, a single one of these draws would dominate the terminal wealth of any lifetime that contained it.

3.5 Market disruptions and the survivorship question

There are 16 contiguous runs of missing data across the countries of the panel, concentrated in the two world wars. The treatment of these gaps is a substantive modelling decision rather than a data-cleaning one. Interpolating across them would smooth away exactly the episodes that motivate the whole exercise; dropping the countries would reintroduce the survivorship bias the panel exists to avoid.

We do neither. Gaps are preserved as gaps, and the bootstrap is constrained to draw only blocks that lie entirely within a contiguous run of available data for the country in question. A country with a six-year wartime hole contributes blocks on either side of it but never across it. The consequence is that the sampler is *less* able to draw from countries with disrupted histories at long block lengths, which we quantify rather than assume: the run-length admissibility statistics are reported in Section 4.1.

4. Methodology

The pipeline that turns the panel into a certainty equivalent is the replicated study's, with one change that matters here: the public pension is a parameter rather than a fixture, so the same simulated lifetimes can be scored under two systems with nothing else moving between them.

4.1 The cross-country stationary block bootstrap

A simulated lifetime is 68 years long and is assembled from contiguous blocks of history. The sampler is a stationary block bootstrap in the sense of Politis and Romano (1994): block lengths are drawn from a geometric distribution with mean 10 years, truncated to [1, 40], and blocks are laid end to end until the horizon is filled.

Two features distinguish it from a textbook block bootstrap and both matter for the result.

- **Blocks are calendar-joint.** A block is a (country, start-year, length) triple, and the same triple is applied to all five series simultaneously. Domestic equity, international equity, bonds, bills and inflation for a given simulated year therefore come from the same real country in the same real year. This is what preserves the cross-asset correlation matrix of Table 3 without any of it being imposed parametrically.
- **Blocks respect data gaps.** Admissibility is enforced through pre-computed run-lengths: for each (country, year) the sampler knows how many consecutive years of complete data follow, and only draws a block that fits inside a run. A wartime hole is never bridged.

Across the production run the sampler drew 30,848 blocks with a mean realised length of 8.82 years against a target of 10, a median of 6, a ninetieth percentile of 20 and a maximum of 40. The realised mean falls short of the target for a mechanical reason worth stating: truncation at the maximum and the gap-admissibility constraint both bite from above, so the effective block length is always slightly below the nominal one.

The country for a lifetime is drawn once, with probability proportional to the length of that country's usable history. Both choices are varied in Appendix C of the companion study: redrawing the country at every block, and weighting countries uniformly, both leave the ranking unchanged. The weighting choice has little room to

bite here, because every country carries between 115 and 131 usable years — history weighting and uniform weighting assign nearly the same probabilities. The choice would matter on a panel with short histories, where a country covering only one era could otherwise speak as loudly as one covering a century and a half.

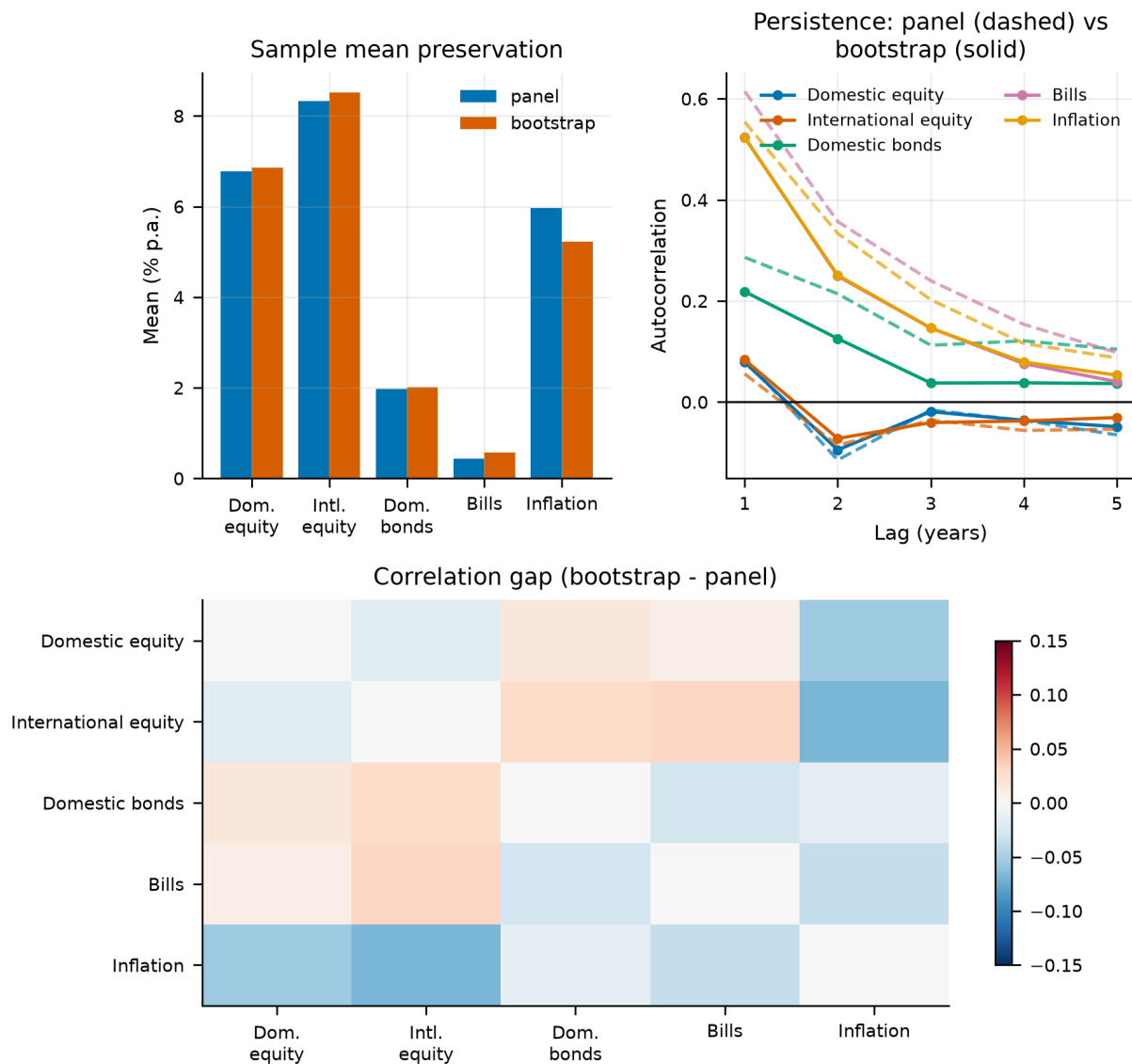
Table 5. Does the bootstrap reproduce the panel it samples from?

Series	Bootstrap mean	Panel mean	Gap (bp)	Bootstrap s.d.	Panel s.d.	s.d. ratio	Bootstrap kurtosis
Domestic equity	6.86%	6.77%	8.3	22.87%	22.50%	1.016	2.9
International equity	8.52%	8.33%	19.0	22.27%	21.69%	1.027	42.5
Bonds	2.02%	1.98%	3.9	11.51%	11.81%	0.974	6.0
Bills	0.58%	0.44%	14.0	7.37%	7.99%	0.922	24.0
Inflation	5.23%	5.97%	-74.1	18.76%	37.66%	0.498	1282.5

Notes. Across the four asset series the means agree to within 19 basis points and the standard deviations to within 8 percent. The inflation row does not: its mean is -74 basis points out and it reproduces 0.50 of the panel's standard deviation, with an excess kurtosis of 1,283. The text below says what that does and does not reach.

The inflation row is the one place the sampler does not reproduce its panel, and it is worth being exact about the reach of that. The mean is 74 basis points low and the standard deviation is 0.50 of the panel's — a shortfall no reader should have to discover from the table. Two things follow, and they point in different directions. The first is that it cannot touch the levels or the orderings reported anywhere in this paper: every series is deflated by its own country's price index in its own year *before* the sampler sees it, so returns, contributions, withdrawals and consumption are real throughout and no quantity in the objective is ever converted from nominal terms. Inflation is carried to preserve the joint draw, not to deflate anything. The second is that one withdrawal rule does read it — the Guyton–Klinger guardrail freezes its nominal payment after a down year, which is a real cut of the realised inflation rate — so that rule's position in the ranking of Section 8 is measured on an inflation series that is too tame. It is not in Section 9's menu and it is not the paper's baseline, so nothing else moves; a study that made the guardrail its subject would have to fix the sampler first. The kurtosis in the thousands is the hyperinflations surviving into the paths, which is the intended behaviour and part of why the variance ratio reads as it does.

The correlation matrix is reproduced to a maximum absolute deviation of 0.031 across the four asset-to-asset pairs that drive portfolio behaviour. The largest deviation anywhere in the matrix is 0.068 and it sits on the inflation row, where the excess kurtosis reported above makes the sample correlation itself unstable. We report both figures rather than the flattering one.

**Figure 2.**

Bootstrap diagnostics. Simulated marginal distributions against the panel they are drawn from, the preservation of the cross-asset correlation structure, and the within-path autocorrelation that block sampling is designed to retain.

Block length and long-horizon dispersion

Block length is the single most consequential tuning parameter in the design, because it controls how much multi-decade persistence survives into a simulated lifetime. The naive expectation is that longer blocks produce more dispersion in long-horizon outcomes. The data say the opposite, and the reason is instructive.

Table 6. Sensitivity of 68-year outcomes to the mean block length

Mean block (years)	Mean annualised	S.d. annualised	5th pct	95th pct	Within-path AR(1)
1	4.41%	3.31%	-1.52%	9.49%	-0.014
5	4.49%	3.27%	-1.41%	9.25%	0.064
10	4.43%	3.20%	-1.53%	8.89%	0.079
20	4.37%	3.14%	-1.69%	8.68%	0.088

Notes. Longer blocks raise the within-path autocorrelation, as intended, but slightly *reduce* the dispersion of annualised 68-year outcomes. Over a horizon this long the historical series mean-reverts, so preserving more of its internal structure preserves that mean reversion too.

Within-path autocorrelation rises monotonically from -0.014 at one-year blocks to 0.088 at twenty-year blocks, confirming that the sampler is doing what block sampling is for. But the standard deviation of annualised 68-year returns *falls* slightly over the same range. This is a long-horizon mean-reversion finding, not a bug: a lifetime that inherits real historical sequences inherits their tendency to revert, whereas one assembled from independent annual draws does not. It is also a caution against reading the block length as a simple dial for "more risk".

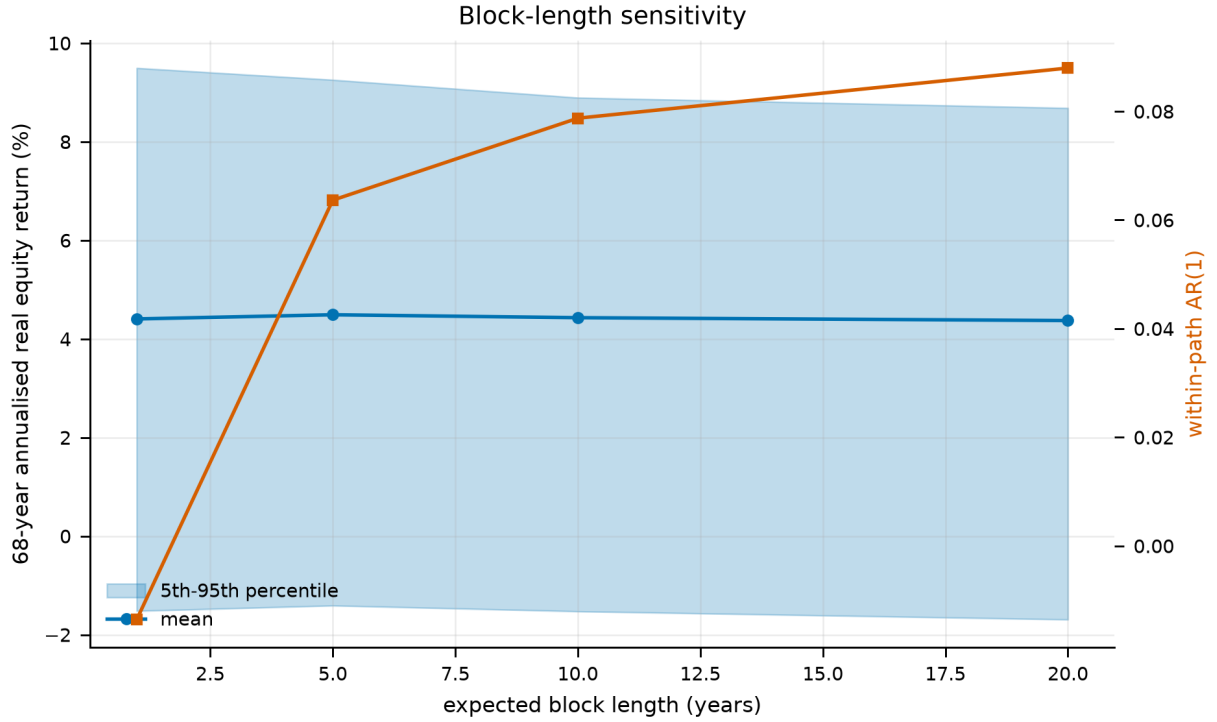


Figure 3.

Block-length sensitivity. Long-horizon dispersion is close to flat in the mean block length even as within-path persistence rises steadily, which is the signature of mean reversion in the underlying series.

4.2 The lifecycle model

A simulated investor begins work at age 25, retires at 63 and dies at 93, giving a 68-year horizon of which 38 are working years. They save a fixed fraction $s = 10\%$ of labour income, rebalance annually to the strategy's target weights, and on retirement switch to a withdrawal rule.

Wealth evolves as

$$W_{h+1} = (W_h + c_h - x_h) \cdot (1 + r_h^p), \quad r_h^p = \sum_a w_{a,h} r_{a,h} \quad (6)$$

where c_h is the contribution in working years and zero afterwards, x_h is the withdrawal in retirement and zero before, $w_{a,h}$ is the strategy's weight on asset a at age h , and all returns are real. Consumption is labour income net of saving while working, and the sum of the social-security benefit and the portfolio withdrawal in retirement:

$$C_h = (1 - s) Y_h \text{ (working)} \quad C_h = B + x_h \text{ (retired)} \quad (7)$$

Labour income follows a deterministic hump-shaped profile ($b_1 = 0.045$, $b_2 = -0.0009$ in age and age squared) multiplied by permanent and transitory shocks with standard deviations 0.10 and 0.25. The permanent shock is a cumulative sum, so income dispersion widens over a career; the transitory shock is i.i.d. The profile is an age effect and carries no economy-wide wage growth; Section 3.6.3 of the companion study measures what that

leaves out and which way it biases the result.

The social-security benefit uses the US primary-insurance-amount bend-point schedule rather than a flat replacement rate, applying 90%, 32% and 15% to successive tranches of career-average earnings. This is not decoration. A flat replacement rate scales with the outcome and therefore insures nothing; a progressive schedule supplies a genuine real consumption floor, which is exactly what determines how much the left tail of a risky strategy actually costs.

Table 7. The six benchmark strategies, as constant weights

Strategy	Domestic eq.	Intl. eq.	Bonds	Bills
50/50 domestic/international equity	50%	50%	—	—
100% domestic equity	100%	—	—	—
100% international equity	—	100%	—	—
60/40 domestic equity/bonds	60%	—	40%	—
100% bills (cash)	—	—	—	100%
Target-date fund (glide path)	see below	see below	see below	see below

Notes. Held constant at every age and rebalanced annually, except the target-date fund.

The target-date fund is the one benchmark whose weights move. Its equity share is piecewise-linear in age through the knots below, interpolated between them; the equity sleeve is split 60/40 domestic/international at every age and the fixed-income sleeve 70/30 bonds/bills. The shape is the industry standard — flat and growth-heavy while young, declining through the decade before retirement, and still declining gently after it.

Table 8. The target-date fund's glide path, at its knots

Age	25	40	55	63 (retire)	75	93
Equity share	90%	90%	65%	50%	35%	30%

Notes. Linear between knots. At age 63 the fund therefore holds 30% domestic equity, 20% international, 35% bonds and 15% bills.

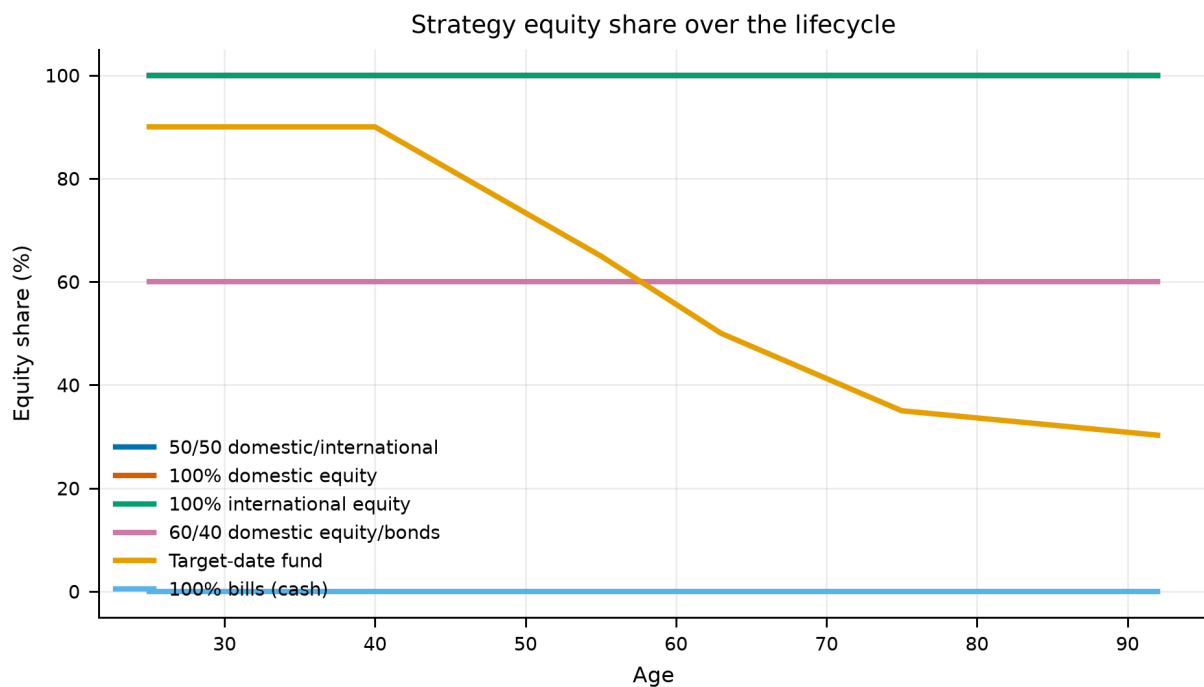


Figure 4.

The six benchmark allocation strategies as equity share by age. The target-date fund is the industry-standard declining glide path; the comparison strategies are held at fixed weights.

4.3 Preferences and the certainty equivalent

Strategies are ranked by certainty-equivalent consumption: the constant, riskless consumption stream that would leave the investor indifferent to the risky one their strategy produces. Under constant relative risk aversion with discount factor β and curvature γ , lifetime utility over the evaluation window is

$$V = \sum_h \beta^h u(C_h) + \beta^H \theta \cdot u(\kappa + W_H), \quad u(c) = c^{1-\gamma} / (1 - \gamma) \quad (8)$$

and the certainty equivalent inverts that utility back into consumption units. We report risk aversions of $\{2, 5, 10\}$ with $\gamma = 5$ as the baseline, $\beta = 0.96$, and a bequest weight $\theta = 2$.

The bequest term uses the shifted-power specification of De Nardi (2004) with $\kappa = 1$. The shift is not cosmetic. Without it, any path that dies with exactly zero wealth contributes $u(0) = -\infty$ at $\gamma \geq 1$, one such path in a hundred thousand collapses the entire certainty equivalent, and the ranking becomes a statement about which strategy avoids exact zeros rather than about which delivers better consumption. This is a genuine trap in lifecycle work and we flag it because we walked into it.

We also report Epstein–Zin preferences in their ex-ante, early-resolution form, which separates risk aversion from the elasticity of intertemporal substitution, with ψ taking the values $\{0.5, 1.5\}$. Under CRRA these two are locked together at $\psi = 1/\gamma$, and a reader is entitled to ask whether the result is being driven by the intertemporal channel rather than the risk channel. Section 6.2 of the companion study shows it is not.

One further specification choice deserves emphasis. Utility is evaluated over the **retirement** window rather than the whole lifetime for the allocation comparisons. The reason is mechanical: with a fixed savings rate, working-life consumption is *identical* across allocation strategies by construction, so including it adds a large constant to every strategy's utility and compresses the differences that the exercise is about. Where a policy *does* change working-life consumption — the retirement-timing and savings-rate studies of Sections 11 to 28 of the companion study — we switch to the whole-lifetime window and say so explicitly.

That is also the answer to a question a reader comparing tables will ask: certainty equivalents in this paper are not all on one scale, and they are not meant to be. Three things vary between sections and each is stated where it applies — the window (retirement-only for the allocation comparisons, whole-lifetime wherever a policy moves working-life consumption), whether the estate is carried, and the path count. Levels are therefore comparable within a table and across tables that share those three, and not otherwise; every comparison this paper draws is within a table for exactly that reason.

4.4 The comparison discipline

Most of the extensions in this paper compare policies that differ in more than one way at once, and the arithmetic of a certainty equivalent will happily credit the wrong difference. Four rules are applied throughout.

- **Matched baselines.** A rule that retires people later is scored against a fixed date matched on the same *mean* retirement age, not against the original age 63. A savings rule that drifts to saving more is scored against a constant rate interpolated at its own realised career average. Without this, a policy is rewarded for working longer or saving more rather than for being smarter.
- **Common random numbers.** Every sweep and every optimiser reuses the same bootstrap draws and the same income shocks, so a difference between two settings is a policy effect and not a sampling artefact. This also makes the objective a deterministic function of the policy, which is what licenses coordinate ascent.
- **Deviation profiles.** Before any structure in a solved schedule is described in words, each element is reset to a neutral reference and the cost measured in basis points. A solved glide path can look highly structured while most of its structure is worth nothing; the profile separates the two.
- **Grid-edge checks.** An optimum sitting on the boundary of its own search grid is a truncation, not an optimum. Every such case is flagged in the text and, where it mattered, the grid was widened and the analysis rerun.

These are not stylistic preferences. Section 28 of the companion study reports a result that fell by roughly sixty percent when a matched baseline replaced an unmatched one, Section 19 of the companion study reports apparent glide-path structure that the deviation profile dissolved, and Section 27 of the companion study reports a functional-form ranking that reversed its interpretation once the grid-edge check was applied.

5. Baseline Results

Before the pension can be shown to reverse the result, the result has to be reproduced. This section is the replication: the all-equity portfolio against the age-declining glide path, under the earnings-related schedule the original study assumes.

All results in this section use 100,000 simulated lifetimes per strategy, drawn from the same bootstrap sample so that strategies face identical market histories. Certainty equivalents are over the retirement window, for the reason given in Section 4.3.

5.1 The headline comparison

Table 9. Headline lifecycle outcomes by strategy

Strategy	CEC $\gamma=2$	CEC $\gamma=5$	CEC $\gamma=10$	P(ruin)	Median wealth at 63	Median cons.	5th pct cons.	Median bequest
50/50 domestic/international equity	1.406	1.048	0.756	11.7%	21.2	1.483	0.707	44.7
100% domestic equity	1.183	0.882	0.668	25.4%	16.1	1.241	0.579	16.7
100% international equity	1.521	1.108	0.782	9.0%	24.4	1.613	0.755	64.2
60/40 domestic equity/bonds	1.105	0.871	0.671	24.2%	13.3	1.149	0.585	10.2
Target-date fund (glide path)	1.214	0.943	0.704	22.2%	16.4	1.273	0.630	9.4
100% bills (cash)	0.847	0.739	0.618	56.4%	6.4	0.856	0.521	0.0

Notes. Consumption and wealth are expressed as multiples of the investor's real income at age 25. P(ruin) is the probability of exhausting the portfolio with retirement years still to fund. 100,000 paths per strategy on common bootstrap draws.

The ordering is unambiguous at every risk aversion tested. At $\gamma = 5$ the 50/50 all-equity portfolio delivers a certainty equivalent of 1.048 against 0.943 for the target-date fund, an advantage of 11.2%. Against the 60/40 portfolio the advantage is 20.3%, and against bills 41.8%.

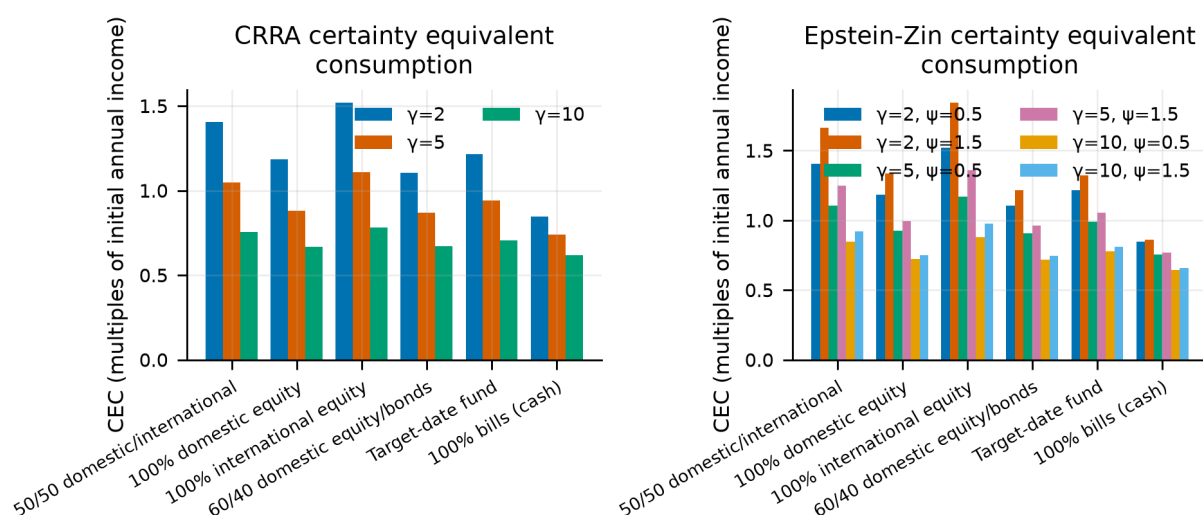
What makes the result more than a restatement of the equity premium is the tail behaviour. The all-equity portfolio has a *lower* ruin probability (11.7%) than the target-date fund (22.2%), a higher fifth-percentile retirement consumption (0.707 against 0.630), and a lower probability of falling below a seventy-percent replacement target (13.6% against 22.6%). The conservative portfolio is not buying downside protection on this panel. It is paying for the appearance of it.

Table 10. Where this implementation departs from the study it replicates, and what each departure does.

What	The replicated study	Here	Does it move the ordering this paper is about?
Return panel	Developed-market panel including proprietary series	Jordà–Schularick–Taylor, 16 markets, openly licensed	No: the all-equity lead over the glide path survives, as the table above reports
International sleeve	Broad global index	Leave-one-out average of the other 15 panel markets	It moves which <i>equity</i> sleeve wins — all-international leads the fifty-fifty split here — and not the equity-against-glide-path ordering
Public pension	United States social security, fixed	A parameter; five regimes	That is this paper's subject rather than a deviation
Horizon	Terminal age, certain	Terminal age in the baseline; survival-weighted alongside it	No: Section 9 scores the whole grid under both aggregations and no sign moves
Withdrawal rule	Fixed real	Fixed real in the baseline; eight rules in the grid	Yes, and that is the paper's finding rather than an artefact of the difference

Notes. The first two rows are data-set differences and the last three are design choices this paper makes deliberately. The sleeve row is the one worth reading closely: a leave-one-out international leg built from sixteen developed markets is a different asset from a global index, and it is why the all-international column wins a comparison the replicated study does not report that way. Nothing in this paper turns on that column.

One caveat on the measure, not the comparison. Both ruin probabilities are aggregated over a horizon that ends at a certain age — here age 93, thirty years after retirement at 63 — which is the convention of the study this paper replicates and the default wherever this one does not say otherwise. A certain horizon overstates the level of ruin for every strategy, because it counts as failure a portfolio exhausted at an age most investors do not reach. Section 8 re-scores the same decisions against a survival curve and puts a size on the overstatement: across the 1,449 combinations of its grid that can run out at all, the fixed horizon reports a median of 2.0 times the survival-weighted number. What a certain horizon does *not* overstate is the *difference* between two portfolios measured the same way on the same draws, which is what this section compares. A reader who wants the level rather than the ordering should take it from Section 8.

**Figure 5.**

Certainty-equivalent consumption by strategy and risk aversion. The ranking is stable in γ ; the gap narrows as risk aversion rises but does not close within the range tested.

5.2 The mechanism is international, not equity

Domestic equity alone is *not* the winning strategy. Its certainty equivalent at $\gamma = 5$ is 0.882, below the 50/50 portfolio's 1.048, and its ruin probability is 25.4% — nearly double the diversified portfolio's. Pure international equity does better still (1.108), which is a consequence of the leave-one-out construction rather than a recommendation: an investor holding "international" equity in this model holds a 15-country average, and no single real investor has that opportunity set without also holding their own market.

This is the one place the replication does not reproduce ACO's headline. Their recommended portfolio is an even 50/50 split, held for life; on this panel that split is beaten by the international leg alone. We read the difference as a property of a 16-country panel rather than a correction to them. The domestic leg here is a single draw from a set that includes several century-long underperformers, while the international leg averages away exactly that risk; with thirty-nine countries and a tradeable international index the two legs are far closer in character, and the case for holding both is correspondingly stronger. Nothing measurable here contradicts the 50/50 recommendation — the narrower claim this panel supports is that a diversified sleeve dominates a concentrated one.

The practical reading is that the case for equities here is a case for *diversified* equities. A single national market can and did deliver multi-decade real losses; the cross-section of national markets did not do so simultaneously. This is a claim about the covariance structure of the panel, and it is the reason the exercise needs an international sample rather than a longer US one.

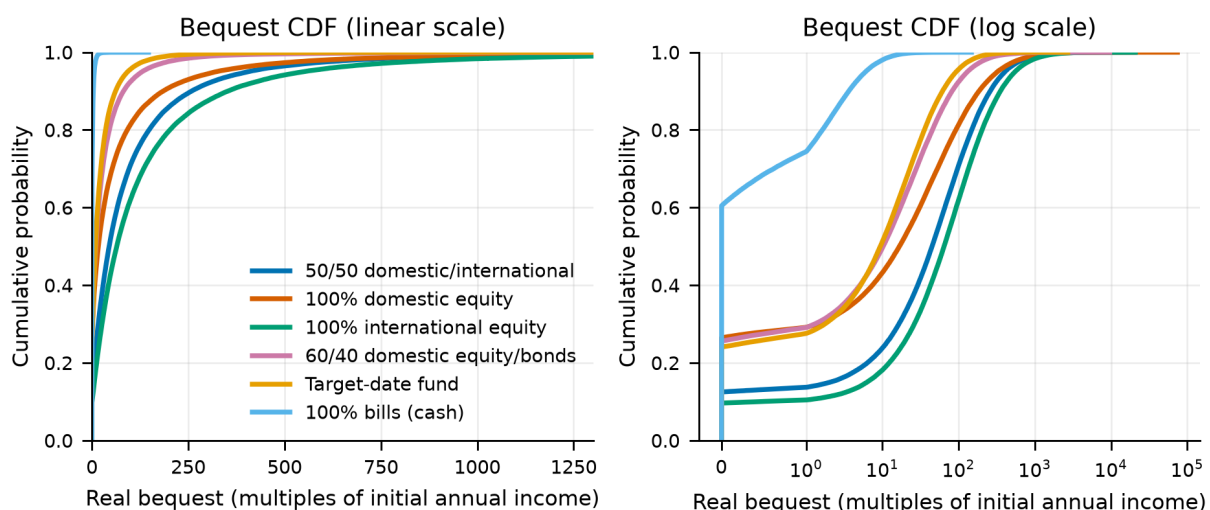


Figure 6.

Cumulative distribution of wealth at retirement. The all-equity distributions stochastically dominate the conservative ones over almost the entire range, including the left tail where the conservative case is usually made.

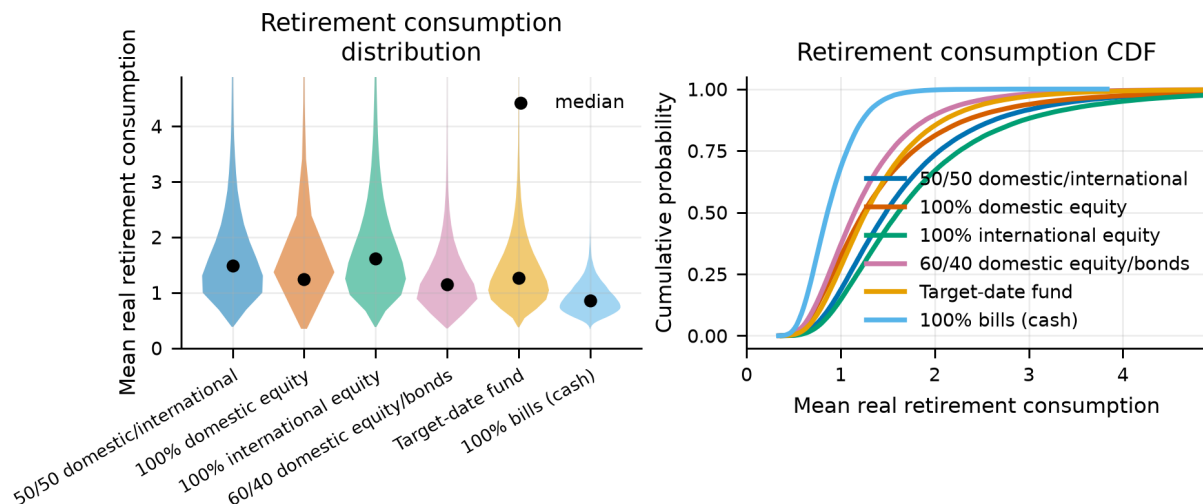


Figure 7.

Distribution of average real retirement consumption by strategy. The vertical line marks the seventy-percent replacement target used in the shortfall statistics.

5.3 Sustainable withdrawal rates

The four-percent rule is the most widely used number in retirement planning. It comes from US historical sequences and it does not survive this panel.

Table 11. Sustainable withdrawal rates on the international panel

Strategy	Sustainable rate at 5% ruin	Ruin probability at 4%
100% international equity	3.48%	9.1%
50/50 domestic/international equity	3.17%	11.7%
Target-date fund (glide path)	2.10%	22.0%
60/40 domestic equity/bonds	1.69%	24.0%
100% domestic equity	1.57%	25.2%
100% bills (cash)	1.07%	56.0%

Notes. The sustainable rate is the highest initial real withdrawal rate, inflation-adjusted thereafter, that leaves at most a five-percent probability of exhausting the portfolio over a 30-year retirement.

At a five-percent ruin tolerance the sustainable rate is 3.17% for the all-equity portfolio and 2.10% for the target-date fund. Withdrawing four percent produces a ruin probability of 11.7% and 22.0% respectively. Two things follow. First, the ordering of strategies is the same here as everywhere else in the paper — the equity portfolio sustains a higher rate, not a lower one. Second, the level is a serious warning: on a sample that includes the twentieth century as most countries lived it, four percent is roughly a one-in-seven proposition even on the best strategy tested.

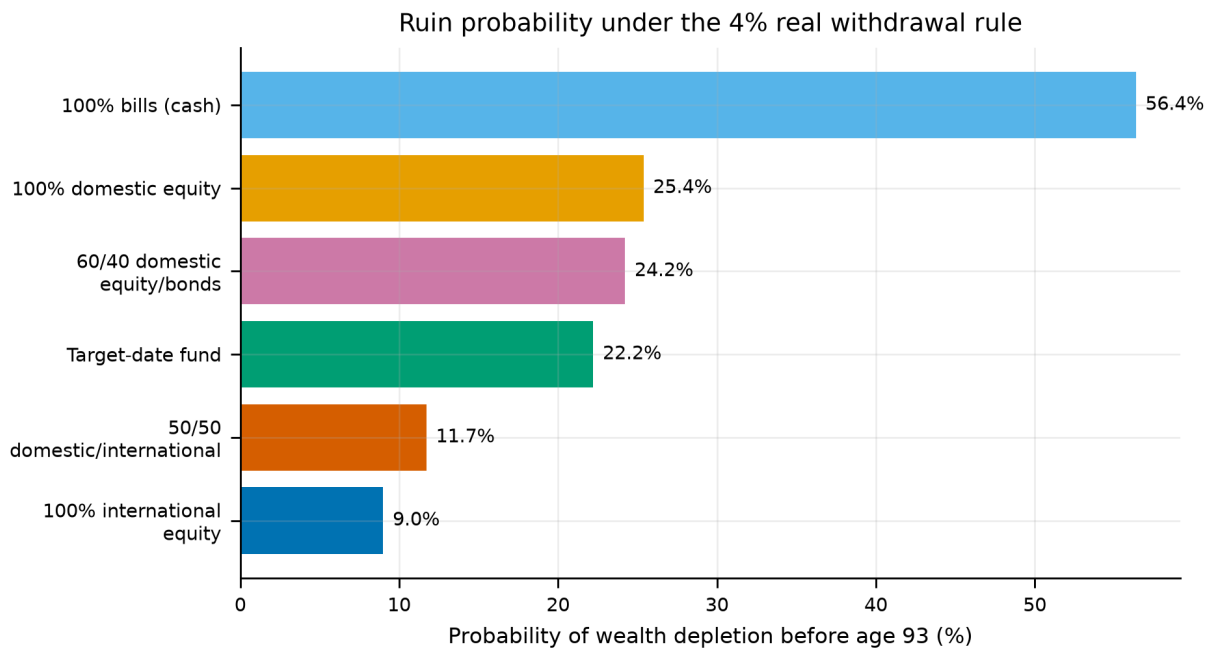


Figure 8.

Probability of portfolio exhaustion as a function of the initial withdrawal rate. The horizontal line is the five-percent tolerance used to define the sustainable rate.

6. The Pension as a Parameter

Every result to this point pays one public pension: the United States' earnings-related schedule, paid in full whatever the retiree owns. That is the assumption the literature this paper replicates makes throughout, and it is the shape most flattering to its conclusion — an unconditional real annuity puts a floor under retirement consumption, which is exactly what makes an equity tail affordable. This section makes the pension a parameter instead of a fixture, and calibrates the settings every later section is scored under.

The regimes are calibrated to a real means-tested schedule because a test invented for the purpose would be a test chosen to produce an answer. Australia's Age Pension supplies the thresholds, the taper and the rate; roughly a third of OECD members pay a benefit of the same shape, withdrawn against the retiree's own assets or income. What the paper claims is about that shape rather than about that country, and the section below is the arithmetic of the calibration rather than a study of one pension system. What each regime does to the portfolio ordering is Section 9, which is where the comparison this paper is about is made.

Table 12. The five settings of the pension parameter. Every later section scores the same simulated lifetimes under each of them.

Regime	Benefit formula	Contribution
Earnings-related, 10% saving	Earnings-related, paid in full	10% voluntary
Earnings-related, 20.2% saving	Earnings-related, paid in full	10% voluntary + 12% mandated, 20.2% net of contributions tax
Means-tested, 10% saving	Flat rate, withdrawn at 8% of each assessable dollar a year	10% voluntary
Means-tested, 20.2% saving	Flat rate, withdrawn at 8% of each assessable dollar a year	10% voluntary + 12% mandated, 20.2% net of contributions tax
Flat rate, no means test	Flat rate, paid in full	10% voluntary

Notes. The first four rows are the design: the benefit formula crossed with the contribution rate, so that no comparison between them moves both at once. The fourth is the combination Australia legislates, and is reported throughout as the confounded case the other three exist to separate rather than as a result in its own right. The fifth switches the means test off at the baseline contribution rate, which separates the flat pension's lower level from the taper that withdraws it. Thresholds, taper and rate are calibrated below; what each setting does to the portfolio ordering is Section 9.

6.1 Calibration

Table 13. The Age Pension for a single retiree, March 2026 indexation, with assets-test thresholds from July 2026, against Australian average weekly ordinary time earnings of \$2,051.10 (ABS, November 2025).

Quantity	Statutory	In average earnings
Maximum single rate	\$1,200.90 a fortnight	0.293
Assets-test free area (homeowner)	\$321,500	3.014
Assets-test free area (non-homeowner)	\$600,000	5.625
Taper	\$3 a fortnight per \$1,000	0.078 a year
Cut-out (homeowner)	\$722,000	6.767
Superannuation Guarantee	12% of earnings, taxed 15% on entry	0.102 reaching the fund
Eligibility age	67, for anyone born after 1 January 1957	—
Paid before that age	60% of the means-tested rate	0.176 at the full rate

Notes. Rates are held as multiples of economy-wide average earnings so the schedule travels across the panel's currencies unchanged. The last row is not a statutory quantity: it stands in for the income-tested unemployment payment a retiree who stops before the eligibility age would claim, and it is a judgement rather than a measurement. Section 33.5 of the companion study sweeps it from nothing to the full rate, because a floor chosen by hand in a study about floors has to be.

The guarantee is a second contribution stream, not a larger first one. The worker still saves this paper's own 10% out of take-home pay; the employer pays 12% on top, 15% of it is taken as contributions tax, and the remaining 10.2% is invested in the same strategy and assessed by the same means test. Total contributions are 20.2% of income against the American saver's 10%. Because both pots hold the same strategy and face no differential tax on earnings in this model, they are financially one pot and are simulated as one; the guarantee supplies 50% of everything contributed.

Working-life consumption is unchanged by the guarantee here, because its statutory incidence is on the employer. If the true incidence is on workers through lower wages — which is what most of the empirical literature finds — a worker under the mandate is paying for it in forgone pay, and every comparison that scores the mandate against a voluntary-saving regime is generous to them by that amount. It does not touch the certainty equivalents, because the utility window in this paper is retirement only. It does mean this is a comparison of *systems as legislated* rather than of two workers with the same lifetime resources, which is why the matched-contribution rows are there.

The taper is the other mechanism, and it is steep. A dollar of assessable assets inside the tapered band costs 7.8% of pension every year it is held. No asset in this panel earns that reliably in real terms, so inside the band the marginal return on wealth is negative — and run backwards, a retiree whose portfolio falls is met by a

pension that rises. A means test is a floor and a ceiling at once, which is why it is assessed on assets as they stand, year by year, rather than settled once at retirement.

7. Who Pays, and Whom the Test Binds

Two objections stand against everything above, and they look like one. The first is that the Superannuation Guarantee has been free: its statutory incidence is on the employer, so the mandated arms have been handed a tenth of income the voluntary ones never got, and most of the empirical literature puts the economic incidence on wages instead. The second is that the household simulated here retires far past the assets test — on 36.4 times average earnings at the all-equity portfolio this sweep holds, against a cut-off of 6.8 — so the means test at the centre of the paper never binds it.

Three balances at retirement appear in this paper and they are not the same household measured three ways. The first is the matched-contribution household, which saves only this paper's own 10% because holding the contribution rate still is what lets a difference be attributed to the benefit formula. Section 7's carries the Superannuation Guarantee and holds the all-equity portfolio this sweep is run on. Section 1's is the same household at the portfolio the headline compares. Each is stated where it is used; what they share, and all the objection above needs, is that every one of them is several times the cut-off.

The tempting move is to answer both at once: charge the guarantee, watch the balance fall, and let the household walk down onto the test. It does not work, and the reason is the cleanest thing in this section. Under full economic incidence the worker funds the contribution out of wages, so take-home pay falls; but the contribution is still made, so the fund receives the same money and compounds it identically. **Incidence changes what the household gave up, not what it arrives with.** The balance at the pension age is the same to the last cent at either end of the dial, and the two objections need two answers.

7.1 Charging the guarantee

The first dial charges a share of the employer contribution against wages, from nothing to all of it, crossed with every allocation: 105 combinations on common random numbers. Charging it in full costs 13.4% of lifetime certainty-equivalent consumption and lowers mean consumption during the working years by 14.8%, which is the transfer itself. That is the size of the free lunch, and the honest reading of every cross-system comparison above is that it is bracketed by the two ends of this dial rather than pinned at either.

Measured the way the rest of this paper measures — a certainty equivalent over the retirement window — the answer is exactly zero, 0.7701 at both ends to every digit. The retiree's problem is untouched by who paid for the balance: same wealth, same pension, same returns. And the allocation does not move either: 100% equity is optimal whether the guarantee is free or funded out of forty years of wages. What the free guarantee inflated was the level of consumption under the mandate, not the portfolio chosen to deliver it.

7.2 A household the test does bind

The second dial moves the balance directly. It scales what reaches the pension age, applied at the retirement boundary and nowhere else, so the career is identical across the grid and a change in the retiree's allocation is a response to the balance rather than to what was given up for it. It is a counterfactual and worth being blunt about: not a claim that real households save less than this one, but the observation that a household near the assets test holds a fraction of what this one holds, and that the retiree's problem at that balance is the one the model in Section 2 speaks to. Nothing near the test is reachable otherwise.

Table 14. The equity share a retiree wants, by where the balance leaves them against the assets test, under two withdrawal rules. Balances are multiples of economy-wide average earnings; the free area is 3.01 and the cut-off 6.77. The pension begins the day work stops in both columns, so every retirement year is under the test.

Median balance at retirement	Position	Wanted equity, fixed real rule	Wanted equity, share of balance
0.72×	below the free area	0%	100%
1.45×	below the free area	0%	100%
2.42×	below the free area	0%	100%
3.38×	inside the taper band	0%	100%
4.35×	inside the taper band	0%	100%
5.31×	inside the taper band	0%	100%
6.28×	inside the taper band	0%	100%
7.25×	above the cut-off	0%	100%
8.21×	above the cut-off	0%	100%
9.66×	above the cut-off	0%	100%
12.08×	above the cut-off	0%	100%
16.91×	above the cut-off	0%	100%
28.98×	above the cut-off	10%	100%
48.30×	above the cut-off	55%	100%

Notes. Each row is the best of twenty-one equity shares at that balance, scored on certainty-equivalent retirement consumption over the same simulated lifetimes.

Under a fixed real withdrawal the prediction fails, and it fails in the direction the wealth-tax reading would have picked. Wanted equity is 0% below the free area, 0% inside the band and 55% above the cut-off — monotone in wealth, with the minimum where the model said the maximum should be. A household the means test actually binds wants no equity at all.

The reason is the withdrawal rule, and it is worth stating precisely because it is the most useful thing in this section. A fixed real rule spends a set amount and never spends what the portfolio earns. For a household near the assets test that is close to the worst available arrangement: a good equity outcome converts into assessable assets, the pension is withdrawn against them at 7.8% a year, and consumption does not rise by a cent. The upside is confiscated and the downside is not. Equity is then dominated, and no amount of insurance in the shape of the budget line can rescue it.

Change the rule and the answer changes completely. Spending a share of the *current* balance instead — so the gain is consumed as it arrives rather than assessed — the same household wants at least 100% equity at every position, below the free area, inside the band and past the cut-off alike — at least, because that is the top of this grid and Section 7.5 shows the household would take more if it were offered. The swing between the two rules averages 95 percentage points, which is the whole allocation. So the portfolio a means-tested retiree should hold is not a fact about the means test. It is a fact about the means test and the drawdown rule together — and the rule is the half the retiree can choose.

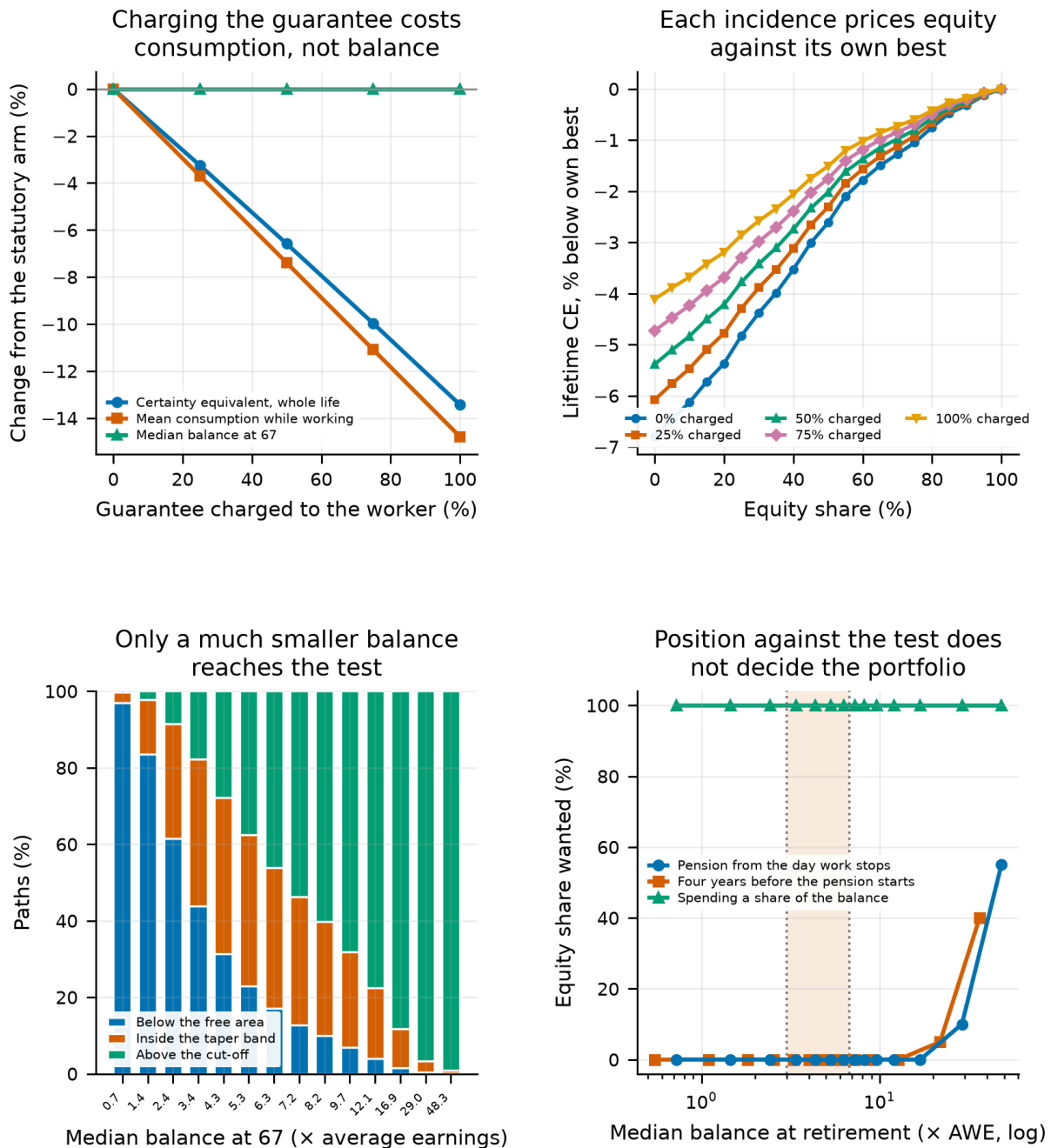


Figure 9.

Two dials on the mandated arm. Charging the Superannuation Guarantee to wages (top row) moves lifetime consumption and leaves the balance untouched; moving the balance (bottom row) is the only way to put a household on the assets test. The lower right panel is the model's prediction, tested.

7.3 What the panel costs this corner

The two results above are checked against the preference specification and not against the panel, which is the objection Section 9 answers for its own numbers. So the balance sweep is recomputed 16 times, once with each country's history removed, and the equity share the retiree wants is read off each one.

The answer is 0% equity in every one of the sixteen deletions. Not close to 0% — identical, to every digit the sweep resolves. A jackknife standard error would be exactly zero and the interval a point, which is why we report the deletions rather than an interval computed from them. That is what a corner solution looks like when it

is checked: the objective is not nearly best at the boundary, it is pinned there, and no fifteen-country subsample of this panel moves it. It is also the reason to state the finding as a direction rather than a number. The panel resolves *which corner* with no uncertainty at all and says nothing about how far inside the boundary a real retiree, facing a real assets test rather than this stylisation, would land.

7.4 And the corner against the objective

The deletions above vary the panel. They do not touch the second objection, which Section 2.2 raises against this result rather than waiting for a reader to: a corner at zero equity is the kind of answer a constant-relative-risk-aversion objective produces when the objective, not the household, cannot tolerate the left tail. Scoring an outcome again costs nothing beside producing it, so the same balance sweep is re-read under five objectives: the baseline, two other risk aversions, and two consumption floors below which the aggregator is not allowed to look.

Table 15. The equity share a means-tested retiree wants under a fixed real withdrawal, by position against the assets test, under five objectives.

Scored under	Below the free area	Inside the taper band	Above the cut-off
the baseline objective	0%	0%	0%
$\gamma = 2$	0%	0%	0%
$\gamma = 10$	0%	0%	0%
a consumption floor of 0.01	0%	0%	0%
a consumption floor of 0.05	0%	0%	0%

Notes. A floor replaces consumption below it before the felicity function is applied, which is what removes the unboundedness the corner would otherwise be exploiting.

The corner is not the aggregator's. Every one of the 15 cells is 0%: the answer does not move at half the baseline risk aversion, at twice it, or with the aggregator forbidden to look below a floor. Whatever else is true of a certainty equivalent at this curvature, the finding here is not made of it. The mechanism in the paragraph above — that a fixed real rule converts a good return into assessable assets and never into consumption — is arithmetic on the budget line and does not consult a felicity function, which is what one would expect to survive this and is what does.

7.5 Whether the other corner is a corner at all

The zero is not at a grid edge in any sense that matters — there is nothing below no equity to want, and the deletions above resolve it exactly. The 100% is different, and the difference is easy to miss. It sits at the top of the grid it was chosen from, so it is consistent with a household that wants exactly the whole portfolio in equity and with one that wants half as much again and was never asked. That is not a quibble about a truncated number: the claim Section 2.1.1 makes is about a *shape* across the assets test, and three regions pinned against a shared ceiling cannot be ranked against one another at all.

So the ceiling comes off. The retiree may borrow against the equity sleeve at the realised bill return plus a spread, in retirement only, so the balance they arrive with is the balance this section scaled rather than one borrowing helped build. The sleeve stays all-equity; the dial is how much of it they hold. The household is the one in the right-hand column of Table 14 and not a household built to resemble it — the same specification, the same pension starting the day work stops, the same fourteen balances, differing from the left-hand column in the withdrawal rule and in nothing else. That is what licenses reading the two columns against each other below.

Table 16. How much of the portfolio a means-tested retiree wants in equity once the grid runs past the whole of it, by position against the assets test, at a borrowing spread of zero.

Position against the test	Balances	Wanted holding, median	Lowest	Highest
below the free area	3	2.00×	2.00×	2.00×
inside the taper band	4	1.75×	1.60×	1.75×
above the cut-off	7	1.50×	1.30×	1.60×

Notes. A holding above 1.00× is borrowing against the sleeve. Zero is the cheapest borrowing there is and therefore the hardest test for the unlevered corner; the paragraph after next prices it properly.

The corner was the grid's. At a borrowing spread of zero the retiree wants between 1.30× and 2.00× the portfolio, and never exactly 1.00× at any of the 14 balances. The 100% reported above is a floor on what this household wants, not the amount they want. Everything the previous subsection says about the *difference* between the two withdrawal rules survives that — it is a difference between a corner at zero and a corner at or above the whole portfolio, which is if anything wider than reported — but the level should be read as a bound.

And the shape is not the predicted one. With room above the whole portfolio the three regions do separate — 2.00× below the free area, 1.75× inside the band, 1.50× past the cut-off — so the ceiling was hiding something. But the ordering runs the wrong way for Section 2.1.1, which puts the maximum inside the band. The maximum is below the free area instead, where the pension is paid in full, and wanted equity falls monotonically as the balance rises. What survives of the prediction is its lower end: the minimum does sit above the cut-off, where the pension has been withdrawn entirely. What does not survive is the claim that the taper band is where a retiree most wants equity.

The reading we take from that is the one the one-period model was always weakest on. A larger guaranteed component raises the optimal risky share; below the free area the guarantee is the full pension and inside the band it is less. The insurance the taper supplies against a falling portfolio is real, and it is smaller than the guarantee it is withdrawing. Section 2 gets the direction of the taper's insurance right and its size wrong, and only a simulation with room above the grid could have said so.

The two rules do not merely want different amounts. They want them in opposite directions. Read down the same fourteen balances, wanted equity under a fixed real withdrawal runs 0% at the poorest and 55% at the richest, rising with wealth; under a percentage-of-balance rule it runs 2.00× and 1.30×, falling. The withdrawal rule does not shift the level of a common profile. It reverses the sign of the wealth gradient itself.

That is the sharpest form of this section's claim, and it is worth separating from the swing in levels. Under a rule that spends the balance, a poorer retiree is the one standing on the largest guaranteed floor relative to their portfolio, so they take the most risk — the textbook endowment result, working normally. Under a rule that does not, the same floor is unreachable: the return cannot be eaten, so it becomes assessable assets and the pension is withdrawn against it, and the poorer retiree is the one with most to lose from holding equity at all. Position against a means test does not have a portfolio implication on its own. It has one sign under one rule and the opposite sign under another, and a plan sponsor who sets a glide path without setting a drawdown default has not chosen between them.

And at a price a household would actually pay, most of the borrowing goes away. The spread over the realised bill return is swept as well as the leverage, and the median wanted holding falls with it: 0% → 1.60×, 1% → 1.50×, 2% → 1.40×, 3% → 1.20×, 5% → 1.00×. The sweep does not reach the price at which the last balance stops borrowing: at 5% the median is 1.00× and 13 of 14 balances want exactly the unlevered portfolio, but one still does not. The break-even is therefore above 5% and this paper does not say where — which is the honest form of the result, and enough for the claim being made: the corner reported above is the grid's, and a household paying a real price for leverage is close to the unlevered portfolio rather than far above it.

Two things keep this from overturning the section rather than qualifying it. The borrowing is worth little: against the unlevered portfolio on the same paths it buys a few per cent of certainty-equivalent consumption, so the household is close to indifferent across a wide range and the ranking of the two withdrawal rules is nowhere near it. And the grid is still a grid: 3 of 14 balances want the top of this one too, so what is established is that the optimum lies above the whole portfolio and not how far above. A retiree who cannot borrow — which is most of them — holds 100% equity and the section's advice is unchanged.

7.6 Whose household this is

Every number above is one retiree — a single renter, on this paper's earnings profile, facing one pair of thresholds. The retirees an assets test actually binds are not that household: most own a home, which the test exempts and which therefore raises the balance at which it starts to bite, and many have a partner, who raises the threshold and the payment together. Both are parameters of the schedule rather than of the model, so the objection can be answered rather than only conceded. Section 2.1's Corollary 1 says what the answer should be: the derivation turns on whether the withdrawal rule's payment reads the balance, and no threshold appears anywhere in it, so no threshold should move the sign. That is specific enough to fail.

Table 17. The equity share each kind of retiree wants, under a rule that reads the balance and one that does not, at every position against that household's own thresholds.

Household	Position against its own test	Wanted equity, fixed real	Wanted equity, share of balance
couple, homeowners	below the free area	0%	100%
couple, homeowners	inside the taper band	0%	100%
couple, homeowners	above the cut-off	0%	100%
couple, renters	below the free area	0%	100%
couple, renters	inside the taper band	0%	100%
couple, renters	above the cut-off	0%	100%
single, homeowner	below the free area	0%	100%
single, homeowner	inside the taper band	0%	100%
single, homeowner	above the cut-off	0%	100%
single, renter	below the free area	0%	100%
single, renter	inside the taper band	0%	100%
single, renter	above the cut-off	0%	100%

Notes. The family home is exempt, so a homeowner is assessed against a lower free area than a renter holding the same wealth in the portfolio; a couple is assessed jointly, against a higher free area and for a higher payment. Only those three schedule parameters move between rows: the career, the contributions, the panel and the simulated lifetimes are identical, so a difference across rows is a property of the schedule and not of the household's history. Each row is the median across the balances that leave that household in that position.

The divide is not this household's. Across four kinds of retiree and three positions against each one's own test, the rule that reads the balance wants more equity than the rule that does not in all twelve cells, and the narrowest gap is 100 percentage points — which is the whole allocation. The corollary predicted that and it is worth saying why it had to: a threshold decides *where* a household meets the test, and the corollary is about what happens to a return once it does. Moving the threshold moves the balance at which the answer changes. It does not move the answer.

And it is the same corner. Wherever the test reaches the return — inside the taper band and above the cut-off — the fixed real rule sits at no equity for every kind of retiree, which is what the corollary says a budget line with no preference parameter in it should produce. Below the free area the pension is paid in full, consumption does not move with the return either way, and the corollary predicts nothing; that band is not read as a corner.

One of those thresholds is weaker evidence than the others and the difference should be visible. The single figures are statutory; the couple pair is a ratio applied to them rather than a separately verified schedule, so it is swept rather than trusted — the treatment Section 33 of the companion study of the companion study gives the pre-eligibility payment, for the same reason. Across five ratios, from 1.00 to 2.25 times the single figure — the first being a couple assessed as though single — the divide holds in every one. Nothing in the couple rows depends on the ratio chosen for them.

Two things this does not do. It is not a calibration to the Australian wealth distribution: every household type is run across the same grid of positions against its own test, which answers whether the result holds for a kind of retiree and not how many retirees of that kind there are. The second question is the more useful one for policy and needs household-level data this study does not carry. And a couple here is one portfolio meeting a couple's schedule rather than two people's retirement problem, which is the right comparison for asking whether the threshold moves the answer and the wrong one for anything else.

The rule divide, by the household the test sees

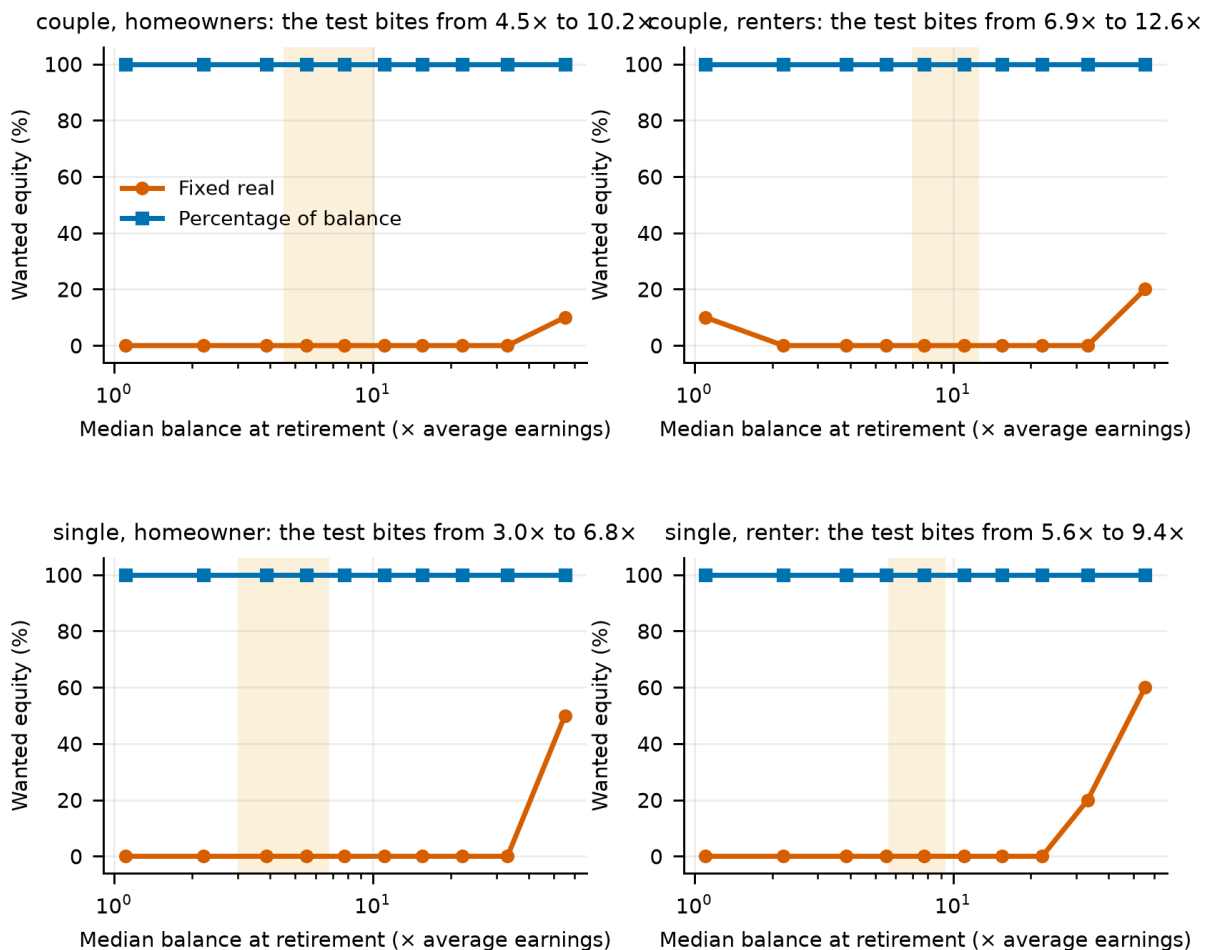


Figure 10.

The rule divide, by the household the assets test sees. One panel per kind of retiree, each carrying its own free area and cut-off as the shaded band, because the same balance sits in different regimes for a renter and a homeowner. Two flat lines a hundred points apart, in four panels whose bands sit in four different places.

This is the same lesson the withdrawal-rule section reaches from the other side, and the two should be read together. There it was a rule that cannot deplete restoring the floor an asset test removed; here it is a rule that cannot spend destroying the case for equity that a floor would otherwise support. Both say the drawdown default

and the portfolio default are one decision, and that a plan sponsor choosing either without the other is choosing in the dark.

8. The Withdrawal Rule a Retiree Should Actually Use

If the mechanism is a missing floor rather than a tax on wealth, a household should be able to build its own floor out of the withdrawal rule. Before that can be tested, the rules have to be compared on an honest footing: every comparison so far scores a retirement that ends on a known date, which hands the rules that amortise to one the answer. This section re-solves the rule, the rate and the allocation together against a real survival curve, and finds which rule a retiree should actually use. What that rule then does to the pension comparison is the next section.

Every comparison of withdrawal rules so far — Section 30 of the companion study's sweep, Section 31 of the companion study's joint optimisation — scores a retirement that ends at ninety-three with certainty. That is not neutral between rules. A rule that sets its level from a planning horizon is handed the answer when the horizon is a known constant, because dividing by the years remaining is very nearly optimal once there is no uncertainty about how many there are. A rule that hedges longevity pays a premium against a risk the model has switched off. And ruin counts as failure a portfolio exhausted at ninety-one for an investor who most likely died some years before. What the fixed horizon flatters, then, is horizon-based rules as a class — and the table below shows it does not pick the same member of that class the survival-weighted objective does, which is a sharper way of putting the same point than naming one rule in advance would have been.

Section 17 of the companion study re-weights the aggregation by a Gompertz survival curve and finds it does not change which *allocation* wins. It also says, in terms, that the treatment is only approximate for the horizon-based rules, “which would themselves change if they knew the mortality table”. This section is that unfinished sentence: the rule and the rate are re-solved alongside the allocation, on one objective.

The grid is 2,280 combinations of allocation, rule and rate, each scored twice on the *same* simulated lifetimes — once over the fixed horizon, once survival-weighted. No path is re-drawn between the two, so a difference between the scores is the aggregation and nothing else. On this calibration the expected age at death is 81.9, so a retiree stopping at 63 should plan for 18.9 years against the 30 assumed throughout. The model was not merely uncertain about the horizon; it was assuming one substantially longer than the median retiree gets.

Table 18. The best combination under each aggregation, $\gamma = 5$.

Scored over	Rule	Equity	Domestic	CEC
fixed horizon	gompertz	100%	10%	1.3844
survival-weighted	amortisation (6% assumed return)	100%	10%	1.4232

Notes. Both winners derive their level from a planning horizon and so set no rate of their own.

The horizon changes the rule and nothing else. Freeing the allocation to be re-chosen for a real lifespan is worth +0.0000% and freeing the rate +0.0000%; freeing the rule is worth +2.81%, which is the whole of the +2.81% available from re-choosing all three, with an interaction of +0.0000%. The first, second and fourth of those are not rounded zeros: the allocation and the rate that win over a fixed horizon are the same ones that win over a real lifespan, so freeing them changes nothing at all. The three decisions do not interact here, and that is worth stating as plainly as an interaction would have been.

The rate optimum is interior: the best rate-setting rule, percentage of balance, wants 8.0% inside a grid running 3.0% to 12.0%, so the curve turns over rather than stopping and the number is a peak rather than the end of the grid.

The assumed return has an interior optimum too. Amortisation (6% assumed return) wants 6% inside a grid running 0% to 12%, so over-assuming does begin to cost and the sweep can see where it starts. This is a dial the rule comparison in Section 30 of the companion study never swept: the amortisation rule enters that menu at assumed returns chosen by hand, and the best of those is not the same thing as an optimum.

The rate a rule wants spans 3.5 percentage points. Percentage of balance wants 8.0% and fixed real wants 4.5%, across 9 rules that set a rate at all. The tempting explanation is that the rules which cannot run out want the high rates, and it is nearly right but not right: endowment (light smoothing) wants 8.0% and *can* deplete, tying the top of the list. What sets the rate is how far spending scales with the portfolio rather than the ability to run out as such — a lightly smoothed endowment rule is most of a percentage of balance, and is priced like one.

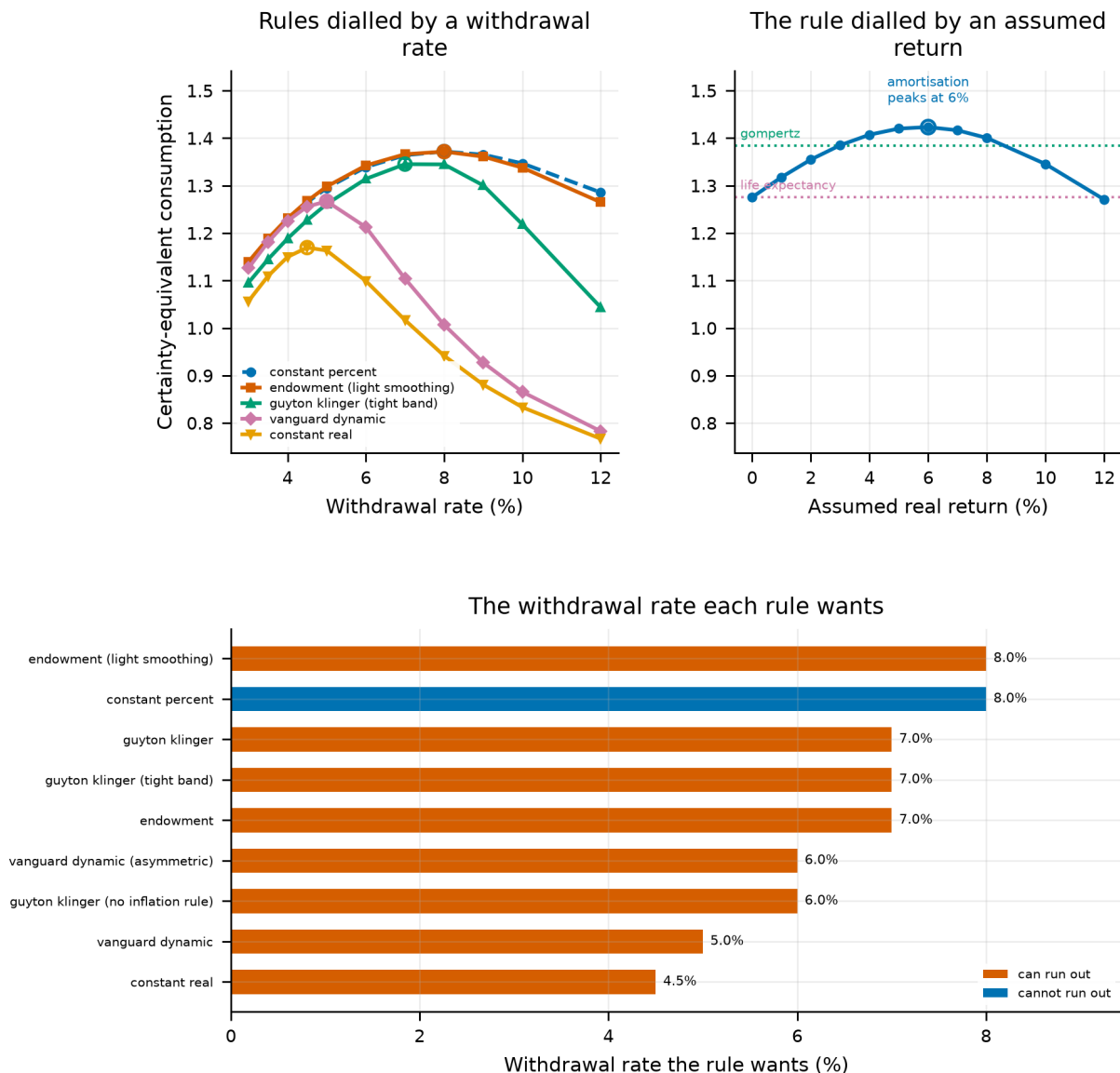


Figure 11.

Where each dial that sets a spending level actually peaks. Top left, the certainty equivalent against the withdrawal rate for the five rules that take one; rings mark where each curve turns over, and solid lines can run out where dashed cannot. Top right, the same against the assumed real return for the rule dialled by one instead, with the two rules that set no dial at all drawn as the constant levels they are. A withdrawal rate and an assumed return are different measures, so they get an x-axis each; the vertical scale is shared between them, which is what lets the two peaks be compared by eye rather than by reading numbers off two ranges. Below, the withdrawal rate each rule wants.

8.1 Which rules a real lifespan promotes

Table 19. Every rule at its own best rate and allocation under the objective being ranked.

Rule	Fixed horizon	Real lifespan	Places gained
amortisation (6% assumed return)	5	1	+4
amortisation (5% assumed return)	3	2	+1
amortisation (7% assumed return)	7	3	+4
amortisation (4% assumed return)	2	4	-2
amortisation (8% assumed return)	11	5	+6
amortisation (3% assumed return)	4	6	-2
Gompertz	1	7	-6
percentage of balance	8	8	+0
endowment (light smoothing)	9	9	+0
endowment	13	10	+3
amortisation (2% assumed return)	6	11	-5
amortisation (10% assumed return)	19	12	+7
Guyton–Klinger (tight band)	14	13	+1
Guyton–Klinger	15	14	+1
Guyton–Klinger (no inflation rule)	18	15	+3
amortisation (1% assumed return)	10	16	-6
Gompertz (+5y buffer)	12	17	-5
Vanguard dynamic (asymmetric)	22	18	+4
amortisation (0% assumed return)	16	19	-3
life expectancy	16	19	-3
amortisation (12% assumed return)	23	21	+2
Vanguard dynamic	21	22	-1
life expectancy (+5y buffer)	20	23	-3
fixed real	24	24	+0

Notes. Scoring each rule at a setting chosen to suit a different horizon would confound the rule with the rate.

Part of the reason is the spending level rather than the mortality table. Rank each rule by the share of the balance it draws in the first retirement year — a number every rule has, however it arrives at one — and that order matches the survival-weighted ranking with a rank correlation of +0.48, against +0.06 under the fixed horizon. The contrast is the finding rather than either level, and at +0.48 it is a contributing cause and not the whole of one. The winning rule has never heard of a survival curve: the best rule that reads one ranks 7. So the answer to “should the withdrawal rule know how long you are likely to live?” is that it does not need to — it needs to spend as though the answer were shorter than thirty years, and there is more than one way to arrange that.

8.2 The ruin number everyone quotes

Ruin roughly halves. On the best rule that can actually run out — endowment — the fixed horizon reports 0.9% against the real lifespan's 0.5%, and across all 1,449 combinations that can deplete the median ratio is 2.0. Nothing about any portfolio changed. What changed is that the earlier number counts as a failure a portfolio exhausted at ninety-one for somebody who, on this survival curve, most likely died before reaching it. Every ruin probability elsewhere in this paper carries the same overstatement.

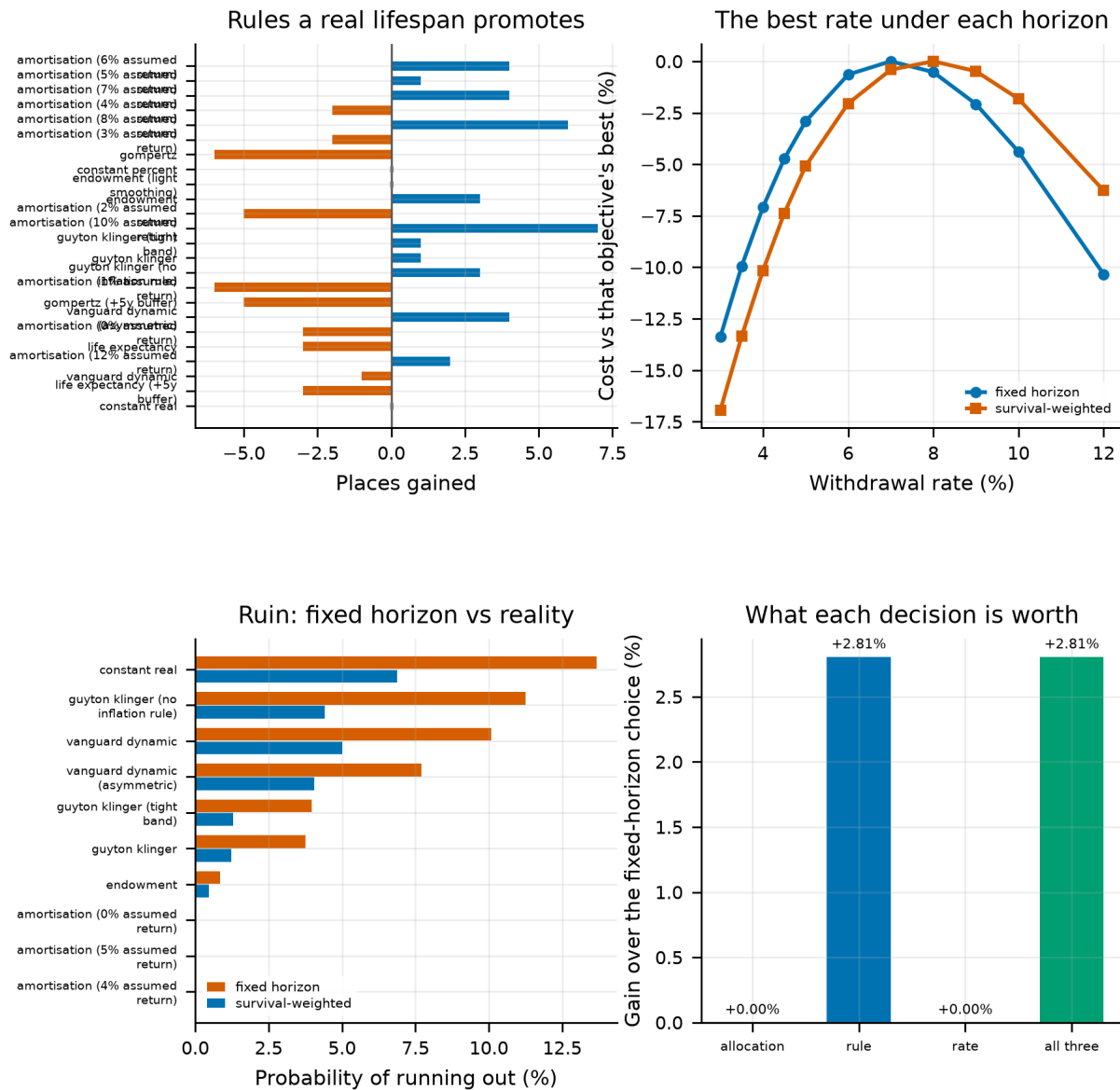


Figure 12.

What an uncertain lifespan does to the rule, the rate and the risk. Top left, how far each rule moves in the ranking; top right, where the best rate sits under each objective, each scaled to its own best so only the shape is compared; bottom left, ruin under both aggregations; bottom right, what re-choosing each decision is worth.

8.3 How firmly the rule is selected

The family is selected far more sharply than the member. At the top of the survival-weighted ranking is amortisation (6% assumed return) at 1.4232, against 1.4201 for amortisation (5% assumed return) — a margin of 0.22%, between two settings of the same rule. Section 9 reads this section's pick as the rule a retiree should use, so the pick had better say how close it is. What the grid resolves is that a horizon-based rule wins; which assumed return inside the family does is not something these differences separate.

Table 20. The rule the survival-weighted objective picks, at each weight on the estate.

Bequest weight	Winning rule	Runner-up	Margin over the runner-up
0	amortisation (6% assumed return)	amortisation (5% assumed return)	0.24%
1	amortisation (6% assumed return)	amortisation (5% assumed return)	0.23%
2	amortisation (6% assumed return)	amortisation (5% assumed return)	0.22%
5	amortisation (6% assumed return)	amortisation (5% assumed return)	0.19%
10	Gompertz	percentage of balance	0.87%

Notes. Each rule is taken at its own best settings under each weight, so a rule is not penalised for a rate that suits a different preference. Nothing is re-simulated: each weight is another score off the same outcomes, exactly as the second objective is.

The pick survives most of the range and not all of it. The winner at the configured weight, amortisation (6% assumed return), also wins from a weight of 0 to 5, so the recommendation is not an artefact of the 2 configured here — which is the check worth running, since the rules in this grid differ in the estate they leave by construction rather than by accident. At 10, a bequest motive stronger than anything this paper assumes, Gompertz takes over, and it does so by spending more slowly: it draws 4.6% of the balance in the first retirement year against 7.3%, so more of the portfolio is still there to be left. That is the right trade once the estate is worth more at the margin than the spending it displaces, and the weight is what decides when it is.

What the weight does not touch is the contrast the next section runs on. Section 9 compares the portfolios twice, once under a rule that can exhaust the account and once under one that cannot, and takes the second from this section's ranking. What it needs from here is therefore which side of that divide the winner falls on, not which member of the winning family it is. The rules winning across the grid are amortisation (6% assumed return) and Gompertz, and every one of them sets its spending from a horizon and so cannot run the account to zero; the fixed real withdrawal, which can, wins at no weight on the grid. The weight moves the member, and at the top of the grid the family, but it never moves the winner across the divide, which is the part of this section Section 9 uses.

What is not modelled: annuities, which are the direct hedge for the risk this section prices and would dominate part of the grid if they were in it; a rule that conditions on realised health rather than on the table it started with; and couples, where the relevant horizon is the second death and the distribution is a different shape. The Gompertz curve is a two-parameter model rather than a life table, and Section 17 of the companion study sweeps those parameters; the grid here is held at that section's central calibration.

9. Which Portfolio Wins, and Under Which Rule

The previous section found the withdrawal rule a retiree should use. This one asks what that rule does to the question the paper opened with, and it is worth being precise about why the question needs asking separately.

Two orderings have been in play. The first is between *portfolios*: all-equity against the target-date fund, which is what Section 5 reports and what the target-date literature is about. The second is between *countries*: whether a household under one pension system consumes more than a household under another, which is what a comparison of certainty equivalents across regimes reports. The companion study reports the second, rule by rule, and it is tempting to read a system moving up that table as the portfolio ordering reversing. It is not the same quantity, and the two can move in opposite directions. This section reports the portfolio ordering directly, under the rule Section 8 selects as well as under the one the literature assumes.

Two things about the grid before the numbers. The menu is five strategies rather than the six of Section 5: cash is dropped, because nothing here turns on it and a portfolio that no reader is choosing between costs a column in every table. And every cell is scored twice — once over the fixed horizon the rest of the paper uses, once

weighted by the survival curve Section 8 argues for. Section 9.6 reports what the second objective does; the tables below are the first, because that is the objective Section 5 reports and the one a reader comparing them will expect.

The rule menu needs the same treatment, because it is eight rules where Section 8 ranks 24. It is not a sample of them. It is the rule the literature assumes and this paper spends by, the amortisation family across every assumed return the grid carries — which is the family that supplies its own floor, and therefore the whole of the mechanism being tested — and one percentage-of-balance rule as the representative of the rules that scale with the portfolio without amortising. Running all 24 against five pension regimes, five portfolios and sixteen deletions would cost a great deal to populate rows nothing in the argument reads. The percentage rule is run at 4% rather than at the 8% Section 8 finds best for it, because here it stands in for the four-per-cent convention the withdrawal-rate literature is written about; at its own best rate it would be a second amortisation-like rule rather than the contrast it is included to provide.

Table 21. The all-equity portfolio's lead over the target-date fund, in certainty-equivalent retirement consumption, for every pension system and every withdrawal rule. Positive means the all-equity portfolio wins.

Withdrawal rule	Earnings-related, 10% saving	Earnings-related, 20.2% saving	Flat rate, no means test	Means-tested, 10% saving	Means-tested, 20.2% saving
fixed real	+11.09%	+15.14%	+12.78%	-13.06%	-2.08%
percentage of balance at 4%	+17.92%	+24.28%	+20.44%	+13.26%	+18.43%
amortisation at 0%	+16.39%	+19.88%	+18.74%	+11.04%	+14.41%
amortisation at 2%	+18.21%	+22.14%	+21.09%	+13.68%	+17.56%
amortisation at 4%	+19.60%	+23.94%	+22.88%	+16.90%	+21.43%
amortisation at 6%	+20.24%	+24.91%	+23.69%	+19.69%	+24.87%
amortisation at 8%	+19.97%	+24.82%	+23.35%	+21.09%	+26.63%
amortisation at 10%	+18.83%	+23.67%	+21.96%	+20.84%	+26.37%

Notes. Every cell is scored on the same simulated lifetimes, so a difference between two cells is the system, the rule or the portfolio and never a different draw of returns. The certainty equivalent is over the retirement window, as everywhere in this paper and for the reason Section 4 gives, so a comparison between two columns saving at different rates carries the retiree's side of that difference and not the worker's; Section 7 prices the other side. The estate is included, because an amortisation rule spends the portfolio to zero by construction and a fixed real rule does not.

The first finding survives being re-derived on this grid. Under a fixed real withdrawal the all-equity portfolio leads by +11.09% under the earnings-related pension and -13.06% under the means-tested one at the same contribution rate, on the regimes Section 6 calibrates and at 50,000 lifetimes rather than the 100,000 of Section 5, which is why the same earnings-related comparison reads 11.2% there and 11.09% here. The two are the same quantity at two Monte Carlo sample sizes and neither is the sampling error that matters, which is the panel's and is measured in Section 9.3.

One caution for a reader who consults the companion study alongside this one. Its pension table has a column headed 'All-intl over 50/50', which compares two *all-equity* portfolios with each other rather than equity against a glide path, and is a question about the international sleeve. Its ranking column is the portfolio comparison, and it agrees with the table above.

And the second finding holds as a statement about portfolios, not only about countries. Under amortisation at 6% — the rule Section 8 selects, which is not the most generous rate in the family and is the one a retiree is being advised to use — the all-equity portfolio leads by +19.69% under the means-tested pension at matched contributions, against -13.06% under the fixed real rule. The family spans +11.04% to +21.09%, so nothing here turns on which rate is picked. Seven of eight rules in the menu return the lead. A rule that supplies its own floor restores the all-equity prescription, and that is a claim about portfolios rather than an inference from one about

countries.

Either way the rule belongs in the statement of the result. Within the means-tested system alone, changing nothing but the withdrawal rule moves the all-equity lead by 34.2 percentage points. A plan sponsor choosing a portfolio default without also choosing a drawdown default is choosing on an axis that explains less than the one they left open.

9.1 Which institution the reversal belongs to

Two of the columns above move two things at once. The regime one country actually legislates is means-tested *and* mandates a contribution, so a comparison between the first column and the last changes the benefit formula and doubles the saving rate in the same step. That is not a detail of the treatment: the larger contribution is what carries this household past the assets test in the first place, as Section 7 reports. The five columns are a 2×2 with a control, and read as one they say something the three-column version could not.

Table 22. The benefit formula crossed with the contribution rate, under a fixed real withdrawal. Each cell is the all-equity portfolio's lead over the target-date fund; the margins are differences along a row and down a column, and the corner is their interaction.

All-equity lead over the target-date fund	Saving 10%	Saving 20.2%	Contribution effect
Earnings-related pension	+11.09%	+15.14%	+4.05 pp
Means-tested pension	-13.06%	-2.08%	+10.99 pp
Pension effect	-24.15 pp	-17.21 pp	+6.94 pp

Notes. Same simulated lifetimes in every cell, and the four are the four regimes of Section 6's menu: the bottom-right cell of the body is the column headed 'Means-tested, 20.2% saving' above, which is the combination Australia legislates, and the top-left is 'Earnings-related, 10% saving', which is the baseline the replicated study assumes.

Held apart, both institutions are large and they work against each other. Replacing an earnings-related pension with a means-tested one, at the same contribution rate, moves the lead -24.15 points. Doubling the contribution rate, under the same means-tested pension, moves it +10.99 the other way. The regime one country legislates does both, and what a single row crossing an earnings-related pension at one contribution rate with a means-tested one at another reports is the residual: -13.16 points, which is smaller than either piece it is made of. The interaction is +6.94 points, and it has the sign the paper's mechanism predicts — compulsory saving is worth more under a means test than under an unconditional pension, because a larger balance is a partial substitute for a floor that has been withdrawn.

The contribution margin is a gross quantity, and the column heading is where to say so. Every certainty equivalent in this paper is taken over the retirement window, for the reason Section 4 gives, so doubling the saving rate enters that margin with its benefit and not its price: forty years of consumption the household did not have. Section 7 prices it on the other measure and the two do not agree, which is the point. Charging the contribution to wages costs 14.8% of mean consumption during the working years and 13.4% of a lifetime certainty equivalent, and moves the retirement-window one by 0.0000% — it cannot move it, because the balance at the pension age is identical either way. So the contribution effect above is what compulsory saving is worth *to the retiree*. That is the quantity a comparison between portfolios needs, and it is not the quantity a comparison between systems as social arrangements would use.

And the reversal, held apart, is a sign this panel can resolve. At matched contributions the means-tested household's all-equity lead is -13.06% with a delete-one standard error of 4.57, an interval of [-22.02, -4.11] that excludes zero, and the sign holds in every one of the sixteen deletions. The cell this paper opened with — the system as legislated, at -2.08% with an interval of [-22.35, +18.20] — is still not resolvable, and now there is a reason for that rather than a shrug. It is what is left when an institutional effect of -24.1 points and one of +11.0 points are allowed to offset inside a single row. The panel can sign the cell those two are measured from; it

cannot sign the remainder.

That is the form the paper's first finding should be stated in. A means test reverses the portfolio ordering under a fixed real withdrawal — measured against a matched household, signed, and robust to dropping any one market. Whether an *Australian* ends up on the wrong side of the ordering depends on a second institution pushing back, and on this panel the two nearly cancel. The first is a statement about pension design and is what the rest of this paper is about; the second is a statement about one country's particular combination, and it is the one that cannot be signed.

9.2 The reversal lives in the worst cell of the table

One row of the table deserves to be read against the rest. Two of the 40 cells are negative, both of them means-tested and both of them in the same row: the fixed real withdrawal — the 4% rule, and the one the lifecycle literature and this paper's earlier sections all spend by. Every other rule in the menu leaves the all-equity portfolio ahead under every pension in the grid. Section 8 has already found that rule is not the one a retiree should use. This table says what it costs the household that does.

Table 23. The all-equity portfolio under each withdrawal rule, under the means-tested pension at matched contributions. The first row is the rule under which the pension reverses the portfolio ordering.

Withdrawal rule	CEC	Ruin, fixed horizon	Ruin, real lifespan
fixed real	0.4130	11.7%	5.9%
percentage of balance at 4%	0.6741	0.0%	0.0%
amortisation at 0%	0.7276	0.0%	0.0%
amortisation at 2%	0.8695	0.0%	0.0%
amortisation at 4%	0.9768	0.0%	0.0%
amortisation at 6%	1.0313	0.0%	0.0%
amortisation at 8%	1.0309	0.0%	0.0%
amortisation at 10%	0.9902	0.0%	0.0%

Notes. Same simulated lifetimes as the table above. Ruin under a fixed horizon is the share of paths whose portfolio is exhausted before the terminal age; under a real lifespan it is the share exhausted before the retiree dies, integrated over the survival curve, which is the measure Section 8 argues for and the smaller of the two by construction.

The fixed real rule leaves this household on 0.4130 against 1.0313 under amortisation at 6% — a reduction of 60% in certainty-equivalent consumption — and exhausts the portfolio on 11.7% of paths against 0.0%. It is dominated on both margins by every other rule in the menu. The ruin margin is the one to state carefully, because Section 8 objects to the measure it is stated in: counting a portfolio exhausted at ninety-one as a failed retirement for a household that most likely died before then overstates what went wrong. Integrated over the survival curve instead, the rule ruins 5.9% of the time against 0.0% — a gap of 5.9 points rather than 11.7. The dominance holds on either measure and is worth about half as much on the better one, which is the honest way to carry it.

That is not a reason to discard the reversal, and we are not discarding it. A fixed real withdrawal is what the withdrawal-rate literature is written about, what most retirement calculators implement, and what the study this paper replicates assumes throughout; a result about the households actually following it is a result about a large number of real people. It is a reason to state the finding with its condition attached. The all-equity prescription survives an asset-tested pension for a retiree who spends a share of the balance, and does not survive it for one who spends a fixed real amount. Which of those a household is doing is a choice, and on this evidence it matters more than the portfolio choice it is usually treated as subordinate to.

9.3 How precisely the panel resolves each sign

Every claim in this paper is a claim about a sign, and the panel is sixteen developed markets whose twentieth centuries were not independent of one another. So each cell above is recomputed sixteen times, once with each country's history removed, and the delete-one jackknife gives the sampling error the panel carries. That is the error to weigh a sign against. Monte Carlo error is not: a hundred thousand paths drive it close to zero without adding a single country of evidence.

Two things about that jackknife are worth stating before its numbers are read, and both cut the same way. Each deletion rebuilds the panel from the markets that remain, so removing one country also changes every other country's international sleeve — it is a leave-one-out average of the rest of the panel, and dropping a market makes it an average of fourteen instead of fifteen. That is the right counterfactual, since the question is what this study would have concluded had the market never been in the data, but it means the sixteen sub-panels overlap in their construction as well as in their history. And the markets themselves are not independent draws: they share the century's wars and depressions, and the effective number of independent long-run episodes is smaller than sixteen. Neither is a reason to trust a standard error built on these deletions without checking it, and the first reason — the skew reported below — is why this section leads with the count of deletions that keep a sign rather than with the interval. Section 9.7 checks the interval itself, by resampling the panel rather than deleting from it, and reports how far the two statements of the same uncertainty differ.

Table 24. Delete-one-country intervals on the all-equity lead, for every system a headline is drawn from.

Pension system	Withdrawal rule	Lead (%)	Jackknife s.e.	95% interval	Sign holds in all 16
Means-tested, 10% saving	fixed real	-13.06	4.57	[-22.02, -4.11]	yes
Means-tested, 10% saving	percentage of balance at 4%	+13.26	3.70	[+6.00, +20.52]	yes
Means-tested, 10% saving	amortisation at 0%	+11.04	2.39	[+6.36, +15.72]	yes
Means-tested, 10% saving	amortisation at 2%	+13.68	2.43	[+8.91, +18.44]	yes
Means-tested, 10% saving	amortisation at 4%	+16.90	2.74	[+11.53, +22.26]	yes
Means-tested, 10% saving	amortisation at 6%	+19.69	3.29	[+13.24, +26.15]	yes
Means-tested, 10% saving	amortisation at 8%	+21.09	3.85	[+13.54, +28.64]	yes
Means-tested, 10% saving	amortisation at 10%	+20.84	4.12	[+12.76, +28.92]	yes
Means-tested, 20.2% saving	fixed real	-2.08	10.35	[-22.35, +18.20]	no
Means-tested, 20.2% saving	percentage of balance at 4%	+18.43	4.76	[+9.10, +27.76]	yes
Means-tested, 20.2% saving	amortisation at 0%	+14.41	3.10	[+8.35, +20.48]	yes
Means-tested, 20.2% saving	amortisation at 2%	+17.56	3.08	[+11.51, +23.60]	yes
Means-tested, 20.2% saving	amortisation at 4%	+21.43	3.27	[+15.01, +27.85]	yes
Means-tested, 20.2% saving	amortisation at 6%	+24.87	3.72	[+17.58, +32.16]	yes
Means-tested, 20.2% saving	amortisation at 8%	+26.63	4.26	[+18.28, +34.98]	yes
Means-tested, 20.2% saving	amortisation at 10%	+26.37	4.66	[+17.24, +35.50]	yes

Pension system	Withdrawal rule	Lead (%)	Jackknife s.e.	95% interval	Sign holds in all 16
Earnings-related, 10% saving	fixed real	+11.09	2.35	[+6.49, +15.68]	yes
Earnings-related, 10% saving	percentage of balance at 4%	+17.92	3.50	[+11.06, +24.78]	yes
Earnings-related, 10% saving	amortisation at 0%	+16.39	2.52	[+11.45, +21.33]	yes
Earnings-related, 10% saving	amortisation at 2%	+18.21	2.92	[+12.49, +23.94]	yes
Earnings-related, 10% saving	amortisation at 4%	+19.60	3.36	[+13.02, +26.19]	yes
Earnings-related, 10% saving	amortisation at 6%	+20.24	3.75	[+12.89, +27.60]	yes
Earnings-related, 10% saving	amortisation at 8%	+19.97	3.98	[+12.17, +27.77]	yes
Earnings-related, 10% saving	amortisation at 10%	+18.83	3.95	[+11.08, +26.57]	yes

Notes. The jackknife standard error over sixteen sub-panels, each holding fifteen markets. An interval containing zero is a cell in which this panel cannot say which portfolio wins.

Table 25. The two cells in which the ordering reverses, on the three statistics the sixteen deletions support.

Cell	Lead (%)	95% interval	Deletions below the point	Implied bias (pp)	Deletions keeping the sign
Means-tested at matched contributions	-13.06	[-22.02, -4.11]	4 of 16	+11.3	16 of 16
The system as legislated	-2.08	[-22.35, +18.20]	4 of 16	+31.8	8 of 16

Notes. The interval assumes the deletions scatter around the point estimate. The implied bias, $(n-1)$ times the gap between the mean deletion and the point, says whether they do. The count against zero assumes neither.

The two reversals are not equally well established, and the statistic that separates them is the last column.

At matched contributions every one of the sixteen fifteen-country panels puts the target-date fund ahead, and the least negative of them is -8.42% — clear of zero rather than close to it. As legislated the sub-panels divide 8–8, which is no evidence about a sign at all.

The middle two columns are the reason to lead with the last one rather than with the interval. A jackknife standard error assumes the deletions sit around the point estimate, roughly half above and half below. In neither of these cells do they: 4 of 16 fall below the matched estimate and 4 of 16 below the legislated one, against seven of sixteen in almost every other cell of the table. The implied biases are +11.3 and +31.8 points. So the intervals we print for these two cells are the weakest of the three statistics, in both, and we print them because withholding an unflattering number is worse than qualifying it.

That leaves the two cells in genuinely different positions, and the difference is not one of degree. The matched cell's claim survives its own diagnostic, because the diagnostic bears on the interval and the claim rests on the count — sixteen of sixteen sub-panels, none of them near zero. The legislated cell's claim has nothing left once the interval is discounted: eight of sixteen, an implied bias 2.8 times the matched cell's, and a mean deletion of +0.04% that does not reverse the ordering at all. We report the first as a finding and the second as a number we cannot sign.

None of this is a property of jackknives. Across the 22 cells in which the ordering does not reverse, the deletions split 7-to-11 around their own point estimates and no implied bias exceeds 3.3 points — the same machinery, the same panel, the same sixteen deletions, well behaved wherever the sign is not close to zero. It is the two cells nearest a sign change that the diagnostic fires on, which is where a jackknife is most likely to be nonlinear and

least likely to be trusted.

Which fifteen-country panels reverse, in the cell where that is in doubt. The question is only live for the system as legislated — at matched contributions every sub-panel reverses and there is nothing to enumerate. Of the sixteen fifteen-country panels built on the legislated arm, eight still put the target-date fund ahead and eight do not: the reversal is reproduced by half the subsamples of the panel it was found in, not by none of them and not by all. The skew the paragraph above reports is real and it is a statement about the estimator, not a licence to say the reversal belongs to the whole panel alone.

Which countries do it is the more useful fact, and the deletions give it directly. Removing Italy moves the cell from -2.08% to +6.22%, a swing of +8.30 points and more than any other market; removing Australia moves it the other way, to -3.54%. The markets whose removal overturns the reversal are the United Kingdom, Italy, Japan, the Netherlands, Norway, Portugal, Sweden and the United States. Two of the largest movers among them are the panel's weakest equity markets over the period, which is the mechanism rather than a coincidence: the all-equity portfolio is hurt more by a century-long domestic underperformance than a glide path is, so dropping one helps it. The reversal is carried by the left tail of the country cross-section, and a panel that happened to exclude Italy or Portugal would not have reported it.

We take three things from that rather than one. The sign is not resolved: 8 of 16 sub-panels overturn it and the interval contains zero. The estimate is not a typical member of its own deletions either, which is what the bias diagnostic says and why the interval is the weakest of the three statistics here. And the reversal has an identifiable source in the cross-section rather than being noise about a point — which is a fact a wider panel could confirm or destroy, and the most useful thing sixteen countries can say about it.

9.4 The interval on the difference, which is the claim

The claim the evidence does support is a difference rather than a level — that changing the withdrawal rule moves the lead by tens of points — and an interval on two levels is not an interval on their difference. The sixteen sub-panels are shared between cells, so the differences are paired and get a jackknife of their own rather than one assembled out of two marginal standard errors.

Table 26. Delete-one-country intervals on the difference between each withdrawal rule and the fixed real rule, under every pension the jackknife covers.

Rule, against the fixed real rule	Earnings-related, 10% saving: difference (pp)	95% interval	Means-tested, 10% saving: difference (pp)	95% interval	Means-tested, 20.2% saving: difference (pp)	95% interval
percentage of balance at 4%	+6.8	[+4.2, +9.5]	+26.3	[+17.1, +35.6]	+20.5	[+0.8, +40.2]
amortisation at 0%	+5.3	[+4.4, +6.2]	+24.1	[+15.4, +32.8]	+16.5	[-2.8, +35.8]
amortisation at 2%	+7.1	[+5.7, +8.6]	+26.7	[+17.7, +35.8]	+19.6	[+0.2, +39.1]
amortisation at 4%	+8.5	[+6.3, +10.7]	+30.0	[+20.4, +39.5]	+23.5	[+3.9, +43.1]
amortisation at 6%	+9.2	[+6.1, +12.2]	+32.8	[+22.7, +42.8]	+26.9	[+7.2, +46.7]
amortisation at 8%	+8.9	[+5.3, +12.4]	+34.2	[+23.8, +44.5]	+28.7	[+8.8, +48.6]
amortisation at 10%	+7.7	[+4.1, +11.3]	+33.9	[+23.5, +44.3]	+28.4	[+8.6, +48.3]

Notes. Paired across the sixteen sub-panels, so the pairing can cancel whatever the deletions do to both cells at once. How much it helps is a property of the cells: where the level is badly resolved the difference built on it inherits some of that.

The rule effect is resolved under both means-tested regimes, and far more tightly at matched contributions. There the standard errors run about 4.9 points and 7 of 7 differences exclude zero; as legislated they run about 10.1 and 6 of 7 do. The gap between those two columns is not a fact about withdrawal rules — the point estimates in them differ by only a few points. It is the badly behaved cell of Section 9.3 propagating into every difference built against it, which is what one should expect and is the reason to print both.

The widest rule effect is amortisation at 8% at +28.7 percentage points, with a standard error of 10.1 and an interval of [+8.8, +48.6]. 6 of 7 differences exclude zero. The pairing helps less than one might hope: the cells correlate 0.27 across deletions at the median, so these intervals are not much tighter than the levels they come from. We report that too, because the reason to compute a difference interval at all is that it *can* be far tighter, and here it is not.

So the rule effect is resolved under both pensions, the reversal is resolved at matched contributions, and what remains unresolved is the legislated cell alone. The honest summary is that sixteen countries can separate a 29-point effect from zero and cannot separate a two-point one — which is what one would expect, and is worth having established rather than assumed.

9.5 And the same grid at other risk aversions

The interval above is what the panel costs in precision. It says nothing about the other objection, which is that a certainty equivalent is a statement about a felicity function as much as about a portfolio. Scoring an outcome again costs nothing next to producing it, so the whole grid is re-scored at risk aversions of 2, 5 and 10. This is the check Section 7 runs against preferences and this section had been running only against the panel; each result is now checked against both.

Table 27. The all-equity portfolio's lead over the target-date fund in the means-tested system at matched contributions, at three risk aversions.

Withdrawal rule	$\gamma = 2$	$\gamma = 5$	$\gamma = 10$
fixed real	+10.71	-13.06	-33.66
percentage of balance at 4%	+32.42	+13.26	+5.93
amortisation at 0%	+30.28	+11.04	+4.36
amortisation at 2%	+31.58	+13.68	+7.01
amortisation at 4%	+32.25	+16.90	+12.83
amortisation at 6%	+32.26	+19.69	+16.03
amortisation at 8%	+31.61	+21.09	+15.83
amortisation at 10%	+30.34	+20.84	+14.40

Notes. The same simulated lifetimes throughout: only the objective they are scored under changes, so the differences across a row are preference effects and nothing else.

One row changes sign, and it is the contested one. Under a fixed real withdrawal the lead runs +10.71% at $\gamma = 2$, -13.06% at $\gamma = 5$ and -33.66% at $\gamma = 10$. Every other row in the table is positive at every risk aversion. So the reversal this paper reports is not a property of the means test on its own, and not of the withdrawal rule on its own: it needs a fixed real withdrawal *and* a household risk-averse enough for the loss of the floor to outweigh the equity premium. Between $\gamma = 2$ and $\gamma = 5$ the sign turns over.

That is a sharper statement of the paper's thesis than the one we set out to make, and a more conditional one. The interaction is not two-way but three-way, and the third term is the one a plan sponsor cannot observe. A default portfolio is chosen for a population, and this table says the right default in a means-tested system depends on where in that population the sponsor thinks the marginal member sits.

9.6 And the same grid on a real lifespan

One objection remains, and it is one this paper raises against itself. Section 8 rejects a retirement that ends at ninety-three with certainty as not neutral *between* withdrawal rules, and re-solves the rule against a survival curve. This section compares portfolios *within* a rule, which is a different exposure — a horizon that flatters the rules dividing by it flatters both portfolios in a cell alike, so the gap between them should be close to insulated even where the levels are not. Should be is not is, so every outcome above was scored a second time under the objective that section argues for.

The levels do move: across the grid the certainty equivalents shift over a span of 15.9 points, from -3.6% to +12.3%. The gaps do not. Across all 40 cells the all-equity lead moves by 1.03 percentage points at the median and 2.28 at the worst, and no sign changes — the dial here is the horizon, and the one that does move a sign is the risk aversion of Section 9.5. The contested cell goes -2.08% to -0.93%. So the insulation is not an argument but a measurement: pricing a real lifespan moves the levels by several times what it moves the comparison this paper is about, and moves no sign at all.

We report it because the alternative was to ask the reader to believe it. A paper that spends a section arguing one objective is the right one, and then reports its central table under another, owes the reader the difference rather than the reasoning.

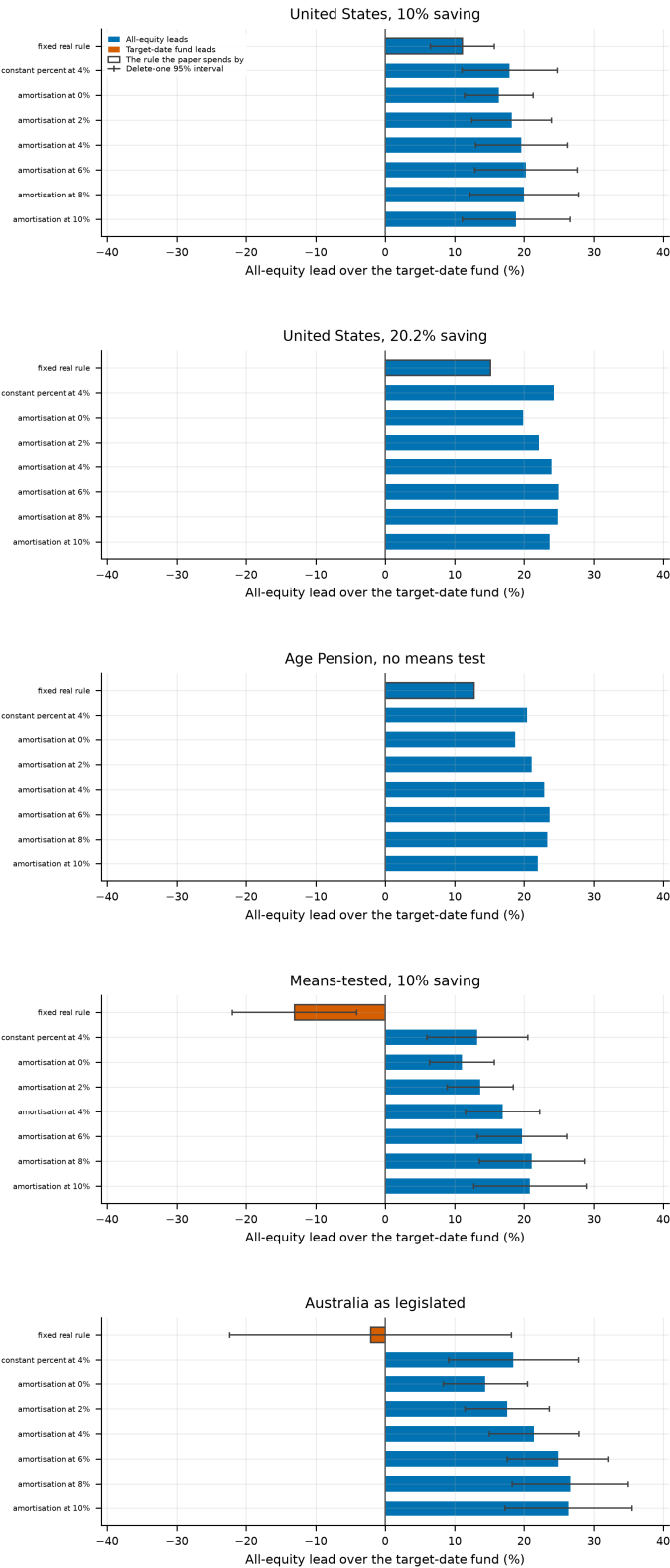


Figure 13. The all-equity portfolio's lead over the target-date fund, by pension system and withdrawal rule. One panel per system on a shared scale; the zero line is the ranking, the outlined bar is the rule the rest of this paper spends by, and the whiskers are delete-one-country 95% intervals where they were computed.

9.7 How much of that precision was the assumption

Section 9.3 reports a delete-one interval and says, before printing it, that the markets are not independent draws and that the deletions overlap in construction. The natural inference is that the standard error is too narrow, and a natural inference is not a measurement. So the panel is resampled instead: 200 times, sixteen markets drawn from the sixteen *with replacement*, the panel and its international sleeve rebuilt from each draw, and the cells the headlines come from re-run on it. The spread across resampled panels assumes nothing about how the deletions scatter and nothing about their independence, because it has no deletions in it.

Table 28. The same cells under two statements of the panel's uncertainty: the delete-one jackknife, and sixteen markets resampled with replacement.

Pension system	Withdrawal rule	Lead (%)	Delete-one s.e.	Resampled s.e.	Ratio	Signed by both?
Means-tested, 10% saving	fixed real	-12.93	4.57	4.78	1.05	yes
Means-tested, 10% saving	amortisation at 6%	+19.54	3.29	2.62	0.79	yes
Means-tested, 20.2% saving	fixed real	-3.90	10.35	6.04	0.58	neither
Means-tested, 20.2% saving	amortisation at 6%	+24.56	3.72	3.03	0.81	yes
Earnings-related, 10% saving	fixed real	+11.00	2.35	1.91	0.81	yes
Earnings-related, 10% saving	amortisation at 6%	+20.15	3.75	3.02	0.80	yes

Notes. Resampled intervals are percentile intervals over the replicates rather than normal approximations, because the normal approximation is the part Section 9.3 distrusts. The point estimate is recomputed on the same cells at the same path count as the replicates, so the ratio is not part Monte Carlo difference. A draw may take a market twice; the multiplicity is carried in how likely that market is to be the one a simulated lifetime lives in, because the panel builder collapses the duplicate.

The concession turns out to have been too pessimistic. The resampled standard error runs 0.58 to 1.05 times the delete-one one, with a median of 0.81. It is not uniformly wider: at the median it is *narrower*, and it exceeds the delete-one figure in one of six cells. The two statements of the panel's uncertainty agree to within about a fifth of themselves. Section 9.3 is right that its deletions are neither independent of one another nor independent in construction; it does not follow that its interval is materially too narrow, and on this panel it is not. We would rather report that than leave a hedge standing that the measurement does not support.

And every sign survives it. No cell the delete-one interval excluded zero for is unsigned once the panel is resampled. The paper's findings therefore do not rest on the assumption the jackknife was making, which is the point of measuring rather than conceding: a concession leaves a reader to guess whether the finding would survive being priced properly, and this says that it does.

The cell the paper leads with, in full: -12.93% with a delete-one standard error of 4.57 and a resampled one of 4.78, a factor of 1.05. It is still signed, and it is signed on the statistic this section has argued for throughout rather than on the one it distrusts.

Two construction choices, checked here rather than cited. How a lifetime's home market is drawn — in proportion to the length of its recorded history, or uniformly — and how the international sleeve is weighted are decisions the panel's design makes, and this paper has been taking the companion study's word that neither matters. Across two arms and twelve cells they move the leads by 1.04 percentage points at the median and 2.73 at the worst, and one sign does not hold: GDP-weighted international sleeve, Means-tested, 20.2% saving under fixed real. That cell is the legislated arm under a fixed real withdrawal — the one this paper has declined to sign since Section 1.1, on an interval containing zero and a count of eight sub-panels in sixteen. A construction choice that moves only the sign the paper already refuses to report is a check that agreed with it, and every cell the paper does sign holds its sign under both choices.

None of this manufactures evidence. Sixteen developed markets resampled sixteen at a time is still sixteen markets, and the resampling inherits every selection effect in what was recorded and openly licensed. What it does is stop the paper quoting an interval it has told the reader not to believe.

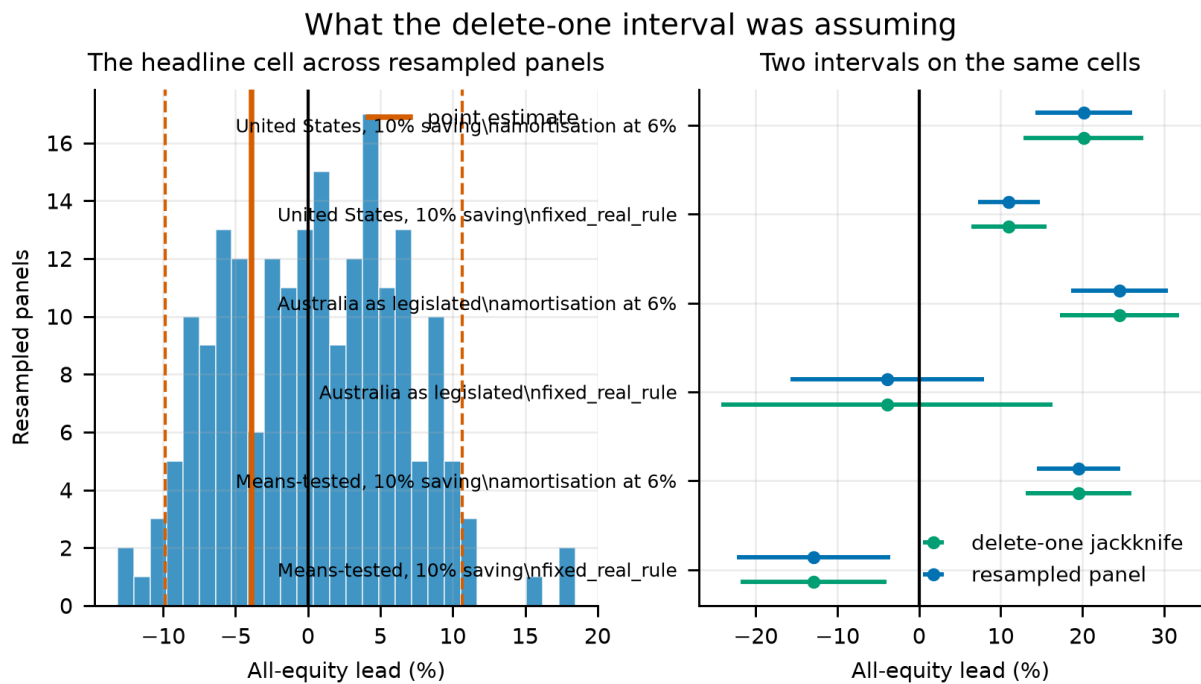


Figure 14.

What the delete-one interval was assuming. Left, the headline cell across resampled panels, with the point estimate and the percentile interval on it. Right, the same cells under both statements of the panel's uncertainty.

10. What Would Change These Conclusions

The results above are the output of a calibrated simulation, and the honest question about any such thing is what would have to be different for it to come out the other way. Four things would.

10.1 The cross-section is sixteen countries

The panel is sixteen developed markets, and they are not sixteen independent draws: equity returns co-move, and the twentieth century happened to all of them at once. Section 9.3 recomputes every cell of the ordering table sixteen times, once with each country's history removed, and the answer is not uniform across the table — which is why this paper separates the institutional effects it can sign from the residual it cannot. The reversal itself, measured against a household matched on contributions, is one of the cells it settles: -13.06% with an interval of [-22.02, -4.11] and the same ordering in all sixteen sub-panels. What it does not settle is Australia's own combination of two institutions that offset. That cell is -2.08% with a delete-one standard error of 10.35 and an interval of [-22.35, +18.20], which contains zero; across the sixteen sub-panels it runs [-3.54, +6.22] and its sign does not survive every deletion. The rest of the table is not like that: 23 of 24 cells have intervals excluding zero, the matched reversal among them. So the cross-section is small enough to leave one sign undetermined and large enough to settle every other, and the two should not be reported in the same voice. And the interval is the weaker half of that, in *both* the cells where the ordering reverses. Only 4 of 16 deletions fall below the legislated estimate, the mean deletion is +0.04%, and the matched cell is skewed the same way — 4 of 16, an implied bias of +11.3 points against the legislated cell's +31.8. So neither estimate is a typical member of its own sub-panels and neither standard error carries what it looks like it carries. What separates the two cells is not the interval but the count of deletions that keep the sign, which assumes nothing about how they scatter: sixteen of sixteen at matched contributions, eight as legislated. That is a statement about the estimator and not about how many countries reverse: eight of 16 fifteen-country panels still put the target-date fund ahead and eight do not, with removing Italy alone moving the cell +8.3 points. Section 9.3 gives the arithmetic; the short form is that the reversal is carried by a handful of identifiable markets in this panel, and no wider panel is available to say whether they are representative. And the delete-one interval is not the only statement of that uncertainty the paper makes. Section 9.7 resamples the sixteen markets with replacement rather than deleting from them, which drops the independence the jackknife assumes; the two agree to within about a fifth of themselves and every sign survives both. That is a check on the method of inference and not on the panel: sixteen markets resampled sixteen at a time is still sixteen markets, and the breadth limitation above is untouched by it. Nothing in this paper should be read as a point estimate, and every claim it makes is a claim about a sign.

10.2 One pension system, stylised

Australia's Age Pension is modelled as an assets test alone: a flat maximum rate, a free area, and a taper, re-assessed every retirement year against the drawn-down balance. Two of the features that separate one real retiree from another under that test are carried: the family home's exemption, which moves a large share of Australian household wealth outside the test and which Section 7.6 runs as its own free area, and a couple's joint assessment, which that section runs as its own threshold and rate. What is not carried is the income test and the deeming rule: the real scheme applies both and pays the lower of the two results, and a household whose income test binds harder than its assets test is solving a different problem from the one modelled here. That changes where a given household sits against the cut-off. It does not change the shape of the budget line in Section 2, which is what the argument turns on — but a reader wanting the number for a particular Australian household will not find it here.

10.3 The household is one household

A single earner, a deterministic wage profile with a permanent shock, no unemployment, no career break, no divorce, no bequest motive beyond a fixed weight, and no home held outside the portfolio. Section 7 moves the balance across the whole width of the assets test, and Section 7.6 crosses that with the two schedule features that most separate real retirees — homeowner against renter, single against couple — finding the rule divide in every cell. Between them those answer the objection that the result was demonstrated on a household the test could not bind, and the objection that it was demonstrated on a household unlike the ones it does bind.

What they do not do is turn one household into a population. Every type in Section 7.6 is run across the same grid of positions against its own test, which says whether the result holds for a kind of retiree and not how many retirees of that kind there are. Calibrating the wealth distribution to Australian household data would answer the second question and needs data this study does not carry. Nor is a couple here two people: the model has one earner, so a couple is one portfolio meeting a couple's schedule, which isolates the threshold and is not a couple's retirement problem.

10.4 No annuity, and no behaviour

The instrument that most directly supplies the floor this paper is about — a life annuity — is not in the choice set, and if it were it would dominate part of the grid. Nor is there any behavioural constraint: the household rebalances annually without hesitation, never panics, and follows its withdrawal rule for thirty years. The rule comparison in Section 8 should be read as what the rules deliver if followed, which is an upper bound on what they achieve in practice.

10.5 What is settled, and what is not

Two things in this paper are robust to all of the above, because they are identities rather than estimates. Charging the Superannuation Guarantee to wages cannot change the balance at retirement, so it cannot change the retiree's problem — that is arithmetic, and the simulation reproduces it to every digit. And a withdrawal rule that spends a share of the balance converts a portfolio gain into consumption, where a fixed real rule converts it into assessable assets; the sign of that difference does not depend on the panel.

Everything else is an estimate. The fuller robustness apparatus — the bootstrap-design sweep, the out-of-sample split, the valuation conditioning, the fee and turnover ladders, the mortality calibration and the parameter sensitivities — is in the companion study, whose Sections 8 and 38 carry it. This paper inherits those results; it does not re-derive them.

11. Conclusion

The case for a fixed all-equity lifecycle portfolio is not one case but a family of them, and which member you are in is settled by two institutions and one preference rather than by the return process. Give an investor an earnings-related pension and a fixed real withdrawal and the all-equity portfolio leads the target-date fund by +11.09%. Change the pension to an asset-tested one, holding the household and its contributions still, and the lead becomes -13.06%: the de-risking glide path, the design the literature spends its length arguing against, wins. Change the withdrawal rule instead and it does not.

That sign the panel can carry: the delete-one interval is [-22.02, -4.11] and the ordering is the same in all sixteen sub-panels. What it cannot carry is the number the country the calibration comes from actually legislates. It pairs the means test with a compulsory contribution, which under that test is worth +11.0 points because a larger balance is a partial substitute for the floor the test removed; the two nearly cancel, the system as legislated lands at -2.08%, and sixteen countries put a standard error of 10.3 points on that remainder — the sign changes when eight of the sixteen markets are removed one at a time. A cross-section of sixteen can sign an institution and cannot sign the difference between two of them that happen to offset.

The other thing the panel resolves comfortably, and in every cell, is the withdrawal rule's effect. Under the system as legislated the rule moves the all-equity lead by 34 percentage points across the menu, and at matched contributions by 34 — larger than either institutional effect under either pension, and an order of magnitude larger than the remainder in dispute. It is the one quantity in this paper that no choice among the arms makes small.

A second check points the same way from a different direction, and it reaches the resolved cell as well as the unresolved one. Re-scoring the identical simulated lifetimes at risk aversions of 2, 5 and 10, the matched means-tested cell runs +10.71% at $\gamma = 2$, -13.06% at $\gamma = 5$ and -33.66% at $\gamma = 10$, and the legislated one +21.91% at $\gamma = 2$, -2.08% at $\gamma = 5$ and -36.40% at $\gamma = 10$ — neither is negative at the lowest of the three — while every other withdrawal rule in the menu stays positive throughout. The reversal therefore needs three things at once: an asset-tested pension, a fixed real withdrawal, and a household risk-averse enough to pay for the floor. Its absence needs only one of the three to fail, and that is as true of the cell the panel signs as of the cell it cannot.

That last sentence is the one we would put first. The reversal holds in one of the eight withdrawal rules we run, and the rule it holds under is the fixed real withdrawal — which costs that same household 60% of its certainty-equivalent consumption against the best rule in the same menu and runs out 5.9 percentage points more often over a real lifespan — 11.7 if ruin is counted to a terminal age nobody is promised. A retiree who follows Section 8's advice never meets the reversal. We report it because it is what the literature's standard assumption produces, not because we think a retiree should be in that cell.

The reason is worth separating from the obvious one. It is tempting to attribute the reversal to the means test's taper, which is steep enough to act as a wealth tax. It cannot be that: this household is past the cut-off before the test applies. It is the loss of the floor, and the cleanest evidence for that is a pair of cells differing in one thing. Paying the calibrated flat rate to this household without testing it leaves the all-equity portfolio ahead by +12.78%; testing that identical benefit, on the same contributions and the same panel, takes it to -13.06%. Nothing else moves between those two runs — not the rate, not the saving rate, not a single simulated return. Withdrawing a pension at the margin and paying a smaller one are different interventions, and only the first reorders portfolios.

The second finding qualifies the first and should travel with it. A withdrawal rule that cannot deplete the portfolio supplies the floor the pension no longer does, and holding everything else fixed it moves the all-equity lead on that same household to +19.69% under the amortisation at 6% rule Section 8 selects, and to +21.09% at the most generous rate in the family, either of which returns the ordering the pension had reversed. So the reversal is not a fact about Australia. It is a fact about any retiree whose income in the bad states depends on the portfolio itself — which includes a saver under an asset test, and excludes one drawing on an annuity, a defined-benefit scheme, or a rule that divides by the years remaining.

The third finding is the one we did not expect and would now put first for a practitioner. Scaling the balance until the assets test genuinely binds — which no household in the sections above does, and which is the objection this design was built to answer — the portfolio the retiree wants stops being a fact about the pension at all. Spending a fixed real amount, they want no equity anywhere the test reaches. Spending a share of the current balance, they want 100% everywhere. The mechanism is not subtle: a rule that never spends what the portfolio earns turns a good return into assessable assets, and the pension is withdrawn against them while consumption stands still. Near a means test, the withdrawal rule does not modify the portfolio decision. It is the portfolio decision.

Three implications follow for default design, and they are implications of a calibrated model rather than costed policy advice. A plan sponsor choosing a glide path is implicitly choosing it against a pension schedule, and on this model the fund that is best under an earnings-related pension need not be the one that is best under a means-tested one. The drawdown default matters as much as the portfolio default: in a system without a floor, supplying one through the withdrawal rule recovers what the pension no longer provides. And for the households a means test actually binds — which are not the wealthy ones the taper is usually discussed around — the two defaults cannot be set separately at all, because the rule decides whether an extra dollar of return reaches consumption or is taken back by the test.

We state these as implications of a calibrated model. It carries no behavioural constraints, prices no annuity, models one household rather than a distribution of them, and — as Section 10 sets out — rests on a sixteen-country panel whose leave-one-out range is wide enough that the direction of these results is far better established than their size. Two of them are exceptions, because they are identities rather than estimates: charging a compulsory contribution to wages cannot change the balance it produces, and a rule that spends a share of the balance converts a portfolio gain into consumption where a fixed real rule converts it into assessable assets. Neither depends on the panel.

References

- Anarkulova, A., Cederburg, S., and O'Doherty, M. S. (2023). "Beyond the Status Quo: A Critical Assessment of Lifecycle Investment Advice." Working paper, University of Arizona. The replication target of this study; every comparison labelled a replication is against this version, and a reader checking the numbers should use the same one.
- Anarkulova, A., Cederburg, S., and O'Doherty, M. S. (2022). "Stocks for the Long Run? Evidence from a Broad Sample of Developed Markets." *Journal of Financial Economics*, 143(1), 409–433.
- Ayres, I., and Nalebuff, B. (2010). *Lifecycle Investing: A New, Safe, and Audacious Way to Improve the Performance of Your Retirement Portfolio*. Basic Books.
- Bateman, H., Eckert, C., Iskhakov, F., Louviere, J., Satchell, S., and Thorp, S. (2018). "Individual Capability and Effort in Retirement Benefit Choice." *Journal of Risk and Insurance*, 85(2), 483–512.
- Bengen, W. P. (1994). "Determining Withdrawal Rates Using Historical Data." *Journal of Financial Planning*, 7(4), 171–180.
- Benzoni, L., Collin-Dufresne, P., and Goldstein, R. S. (2007). "Portfolio Choice over the Life-Cycle when the Stock and Labor Markets are Cointegrated." *Journal of Finance*, 62(5), 2123–2167.
- Bodie, Z., Merton, R. C., and Samuelson, W. F. (1992). "Labor Supply Flexibility and Portfolio Choice in a Life Cycle Model." *Journal of Economic Dynamics and Control*, 16(3–4), 427–449.
- Braun, R. A., Kopecky, K. A., and Koreshkova, T. (2017). "Old, Sick, Alone, and Poor: A Welfare Analysis of Old-Age Social Insurance Programmes." *Review of Economic Studies*, 84(2), 580–612.
- Büttler, M., Peijnenburg, K., and Staubli, S. (2017). "How Much Do Means-Tested Benefits Reduce the Demand for Annuities?" *Journal of Pension Economics and Finance*, 16(4), 419–449.
- Campbell, J. Y., and Viceira, L. M. (2002). *Strategic Asset Allocation: Portfolio Choice for Long-Term Investors*. Oxford University Press.
- Chambers, D., Spaenjers, C., and Steiner, E. (2021). "The Rate of Return on Real Estate: Long-Run Micro-Level Evidence." *Review of Financial Studies*, 34(8), 3572–3607.
- Cocco, J. F., Gomes, F. J., and Maenhout, P. J. (2005). "Consumption and Portfolio Choice over the Life Cycle." *Review of Financial Studies*, 18(2), 491–533. The income profile used here is taken from this estimation.
- Cooley, P. L., Hubbard, C. M., and Walz, D. T. (1998). "Retirement Savings: Choosing a Withdrawal Rate That Is Sustainable." *AAIL Journal*, 20(2), 16–21.
- Dahlquist, M., Setty, O., and Vestman, R. (2018). "On the Asset Allocation of a Default Pension Fund." *Journal of Finance*, 73(4), 1893–1936.
- De Nardi, M. (2004). "Wealth Inequality and Intergenerational Links." *Review of Economic Studies*, 71(3), 743–768.
- Dimson, E., Marsh, P., and Staunton, M. (2002). *Triumph of the Optimists: 101 Years of Global Investment Returns*. Princeton University Press.
- Epstein, L. G., and Zin, S. E. (1989). "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework." *Econometrica*, 57(4), 937–969.
- Fidelity Investments. "Retirement guidelines: how much should I save?" Viewpoints. The $1\times/3\times/6\times/8\times/10\times$ salary-multiple ladder used as an alternative wealth target in Section 27.4 of the companion study.
- Geltner, D. (1991). "Smoothing in Appraisal-Based Returns." *Journal of Real Estate Finance and Economics*, 4(3), 327–345.
- Gomes, F., and Michaelides, A. (2005). "Optimal Life-Cycle Asset Allocation: Understanding the Empirical Evidence." *Journal of Finance*, 60(2), 869–904.
- Guyton, J. T., and Klinger, W. J. (2006). "Decision Rules and Maximum Initial Withdrawal Rates." *Journal of Financial Planning*, 19(3), 48–58.
- Hubbard, R. G., Skinner, J., and Zeldes, S. P. (1995). "Precautionary Saving and Social Insurance." *Journal of Political Economy*, 103(2), 360–399. The mechanism this paper measures in a portfolio setting is the one they identify for saving.

- Iskhakov, F., Thorp, S., and Bateman, H. (2015). "Optimal Annuity Purchases for Australian Retirees." *Economic Record*, 91(293), 139–154.
- Jordà, Ò., Knoll, K., Kuvshinov, D., Schularick, M., and Taylor, A. M. (2019). "The Rate of Return on Everything, 1870–2015." *Quarterly Journal of Economics*, 134(3), 1225–1298.
- Jordà, Ò., Schularick, M., and Taylor, A. M. (2017). "Macrofinancial History and the New Business Cycle Facts." *NBER Macroeconomics Annual*, 31, 213–263.
- Merton, R. C. (1969). "Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case." *Review of Economics and Statistics*, 51(3), 247–257.
- Milevsky, M. A., and Huang, H. (2011). "Spending Retirement on Planet Vulcan: The Impact of Longevity Risk Aversion on Optimal Withdrawal Rates." *Financial Analysts Journal*, 67(2), 45–58. The amortisation rule that wins in Section 8 is theirs.
- Politis, D. N., and Romano, J. P. (1994). "The Stationary Bootstrap." *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Samuelson, P. A. (1969). "Lifetime Portfolio Selection by Dynamic Stochastic Programming." *Review of Economics and Statistics*, 51(3), 239–246.
- Sefton, J., van de Ven, J., and Weale, M. (2008). "Means Testing Retirement Benefits: Fostering Equity or Discouraging Savings?" *Economic Journal*, 118(528), 556–590.
- Viceira, L. M. (2001). "Optimal Portfolio Choice for Long-Horizon Investors with Nontradable Labor Income." *Journal of Finance*, 56(2), 433–470.
- Waring, M. B., and Siegel, L. B. (2015). "The Only Spending Rule Article You Will Ever Need." *Financial Analysts Journal*, 71(1), 91–107.
- Yaari, M. E. (1965). "Uncertain Lifetime, Life Insurance, and the Theory of the Consumer." *Review of Economic Studies*, 32(2), 137–150.